

PREDICTING HOSPITAL READMISSION WITHIN 30 DAYS AMONG PATIENTS WITH DIABETES

Dataset: Diabetes 130-US Hospitals for Years
1999-2008

GitHub Repository: <https://github.com/ran-kalamba/capstone-project>

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Executive Summary

This project developed a binary classification model to identify patients with diabetes who could get readmitted within 30 days of hospital discharge. The analysis used the public Diabetes 130-US Hospitals dataset and constructed a patient-level cohort of 69,973 first eligible encounters, of which 8.97% were followed by an early readmission. Logistic regression, random forest, and XGBoost were compared using Average Precision as the primary metric because of the imbalanced outcome.

XGBoost achieved the strongest test performance, with Average Precision of 0.183, ROC-AUC of 0.653, recall of 0.666, and precision of 0.128 at the training-selected threshold of 0.0788. SHAP analysis indicated that discharge destination, previous inpatient utilization, length of stay, diagnosis group, laboratory activity, medication count, and age contributed to predictions. The model should be interpreted as a risk-ranking tool rather than a causal model.

Fairness analysis found relatively small differences by gender, moderate differences by race, and substantial differences by age. Removing race, gender, and age did not consistently reduce these disparities. The model is therefore suitable only for further prospective evaluation as a tool for offering additional post-discharge support. It should not be used to deny care or determine treatment eligibility.

1. Problem Understanding and Framing

1.1. Business problem and intended use

After patients are discharged at a hospital, some of them are often readmitted after a time-period. For such patients, transition-of-care resources such as discharge counselling, medication reconciliation, early follow-up and care-management calls could be especially beneficial. However these resources are scarce and hospitals need to make informed decisions for which patients should receive these services.

The proposed model is a discharge-time risk-stratification tool: it ranks patients by estimated risk of an inpatient readmission within 30 days so that a hospital can target additional support for such profiles. It is predictive, not causal; a high score does not prove that a particular factor caused readmission or that an intervention will necessarily prevent it.

The prediction timestamp is discharge from the first eligible inpatient encounter. Therefore, only information available by discharge is eligible, including demographics, encounter characteristics, diagnoses, laboratory-test status and results, prior-year utilization, and diabetes medication status or changes. The target is derived from the first subsequent inpatient record captured by the source system.

1.2. Data Science strategy and target

This is supervised binary classification. The source outcome 'readmitted' has three values. Encounters labelled '<30' are mapped to 1; both '>30' and 'NO' are mapped to 0. Thus, the model distinguishes early readmission from every other observed outcome. After cohort construction, there were 6,277 positives and 63,696 negatives among 69,973 patients, a positive prevalence of 8.97%.

Element	Definition
Unit of prediction	One patient represented by the first eligible inpatient encounter.
Positive class	Readmission in less than 30 days ('readmitted = <30').
Negative class	Readmission after 30 days or no recorded readmission ('>30' or 'NO').
Prediction time	At discharge from the index encounter.
Intended decision	Prioritize patients for post-discharge support; not deny care or determine eligibility.

Table 1. Definition of the prediction task, target outcome, prediction time, and intended model use.

1.3. Success Metrics and Business KPIs

Average Precision was the primary model-development metric because only 8.97% of the cohort experienced the positive outcome. Average Precision summarizes performance across all possible decision thresholds by integrating the precision-recall curve. **Therefore it is a reflection of how well the model finds high-risk patients without being misled by the large majority class.**

It was selected over accuracy because even a model that always predicts “no readmission” would still have a 91% accuracy in this context. ROC-AUC is retained as a secondary ranking metric because it can look optimistic under severe imbalance and average precision is more sensitive to performance on the rare positive class.

Threshold-dependent metrics include recall, precision, specificity, F1 score, balanced accuracy, and the confusion matrix. These depend on picking a specific probability cutoff. These metrics inform how many true positives are caught (recall), how many flagged patients are truly high risk (precision), and the trade-offs with false positives/negatives – at the chosen threshold. Brier loss and calibration assess whether predicted probabilities are numerically reliable.

1.4. Ethical and Intended Use Statement

The proposed model is intended to support the allocation of beneficial post-discharge services by estimating the risk of inpatient readmission within 30 days. Potential interventions include discharge counselling, medication reconciliation, early follow-up appointments, and care-management calls. The model is designed to supplement clinical judgement and should not operate as an autonomous decision-making system.

The model must not be used to deny or reduce care, determine insurance eligibility, limit access to treatment, make discharge decisions, or assign blame to patients or clinicians. A low predicted risk must not prevent a patient from receiving support when it is considered clinically appropriate. Similarly, a high predicted risk represents a need for further review rather than proof that readmission will occur.

Predictions should always be reviewed together with the patient’s clinical condition, preferences, social circumstances, and available support. Because the source data are historical and the model showed subgroup performance differences (particularly by age) prospective local validation, human oversight, fairness monitoring, and periodic performance review would be required before any clinical deployment.

2. Data Collection and Understanding

2.1. Source population and provenance

The project uses the public UCI Machine Learning Repository dataset Diabetes 130-US Hospitals for Years 1999-2008 (DOI 10.24432/C5230J). It is a deidentified extract from the Cerner Health Facts database covering ten years of care at 130 US hospitals and integrated delivery networks. The downloadable file contains 101,766 inpatient encounters and 50 columns: two identifiers, 47 features, and the outcome. The source inclusion criteria required an inpatient diabetic encounter, a length of stay of 1-14 days, at least one laboratory test, and medication administration. Consequently, the source is not a sample of all people with diabetes or all hospital admissions.

Primary references:

- UCI dataset page : <https://archive.ics.uci.edu/dataset/296/diabetes+130+us+hospitals+for+years+1999+2008>
- Strack et al. (2014): <https://doi.org/10.1155/2014/781670>

2.2. Original data dictionary

The dictionary below covers every original column used by the CSV. A question mark in the file was parsed as missing. ID-code labels are defined in the companion UCI 'IDS_mapping.csv'; three ID fields are treated as categorical even though they are stored as integers.

Variable	Type	Meaning or unit	Allowed or observed values
encounter_id	Identifier	Unique encounter identifier	Positive integer
patient_nbr	Identifier	Unique patient identifier	Positive integer
race	Nominal	Recorded race	Caucasian, AfricanAmerican, Asian, Hispanic, Other; missing
gender	Nominal	Recorded gender	Female, Male, Unknown/Invalid
age	Ordinal band	Age in 10-year intervals	[0-10) through [90-100)
weight	Ordinal band	Weight in pounds, banded	[0-25), ..., [175-200), >200; missing
admission_type_id	Categorical code	Admission type	UCI mapping; e.g. emergency, urgent, elective, newborn, unavailable
discharge_disposition_id	Categorical code	Discharge destination/status	UCI mapping; e.g. home, transfer, hospice, expired
admission_source_id	Categorical code	Source of admission	UCI mapping; e.g. referral, emergency room, transfer
time_in_hospital	Integer count	Days from admission to discharge	1-14 days
payer_code	Nominal code	Payer category	17 observed labels in file; missing
medical_specialty	Nominal	Admitting physician specialty	72 observed labels in file; missing
num_lab_procedures	Integer count	Laboratory tests during encounter	1-132 observed
num_procedures	Integer count	Non-laboratory procedures	0-6 observed
num_medications	Integer count	Distinct generic medication names	1-81 observed
number_outpatient	Integer count	Outpatient visits in preceding year	0-42 observed
number_emergency	Integer count	Emergency visits in preceding year	0-76 in original; 0-37 in train
number_inpatient	Integer count	Inpatient visits in preceding year	0-21 in original; 0-12 in train

diag_1	ICD-9-CM code	Primary diagnosis, first three digits	Code or missing
diag_2	ICD-9-CM code	Secondary diagnosis, first three digits	Code or missing
diag_3	ICD-9-CM code	Additional secondary diagnosis, first three digits	Code or missing
number_diagnoses	Integer count	Diagnoses recorded in system	1-16 observed
max_glu_serum	Nominal	Maximum serum glucose range or test status	Norm, >200, >300, None/Not measured
A1Cresult	Ordinal nominal	HbA1c range or test status	Norm (<7%), >7, >8, None/Not measured
metformin	Ordinal nominal	Drug prescribed and within-encounter dose status	No, Steady, Up, Down
repaglinide	Ordinal nominal	Drug prescribed and within-encounter dose status	No, Steady, Up, Down
nateglinide	Ordinal nominal	Drug prescribed and within-encounter dose status	No, Steady, Up, Down
chlorpropamide	Ordinal nominal	Drug prescribed and within-encounter dose status	No, Steady, Up, Down
glimepiride	Ordinal nominal	Drug prescribed and within-encounter dose status	No, Steady, Up, Down
acetohehexamide	Ordinal nominal	Drug prescribed and within-encounter dose status	No, Steady, Up, Down
glipizide	Ordinal nominal	Drug prescribed and within-encounter dose status	No, Steady, Up, Down
glyburide	Ordinal nominal	Drug prescribed and within-encounter dose status	No, Steady, Up, Down
tolbutamide	Ordinal nominal	Drug prescribed and within-encounter dose status	No, Steady, Up, Down
pioglitazone	Ordinal nominal	Drug prescribed and within-encounter dose status	No, Steady, Up, Down
rosiglitazone	Ordinal nominal	Drug prescribed and within-encounter dose status	No, Steady, Up, Down
acarbose	Ordinal nominal	Drug prescribed and within-encounter dose status	No, Steady, Up, Down
miglitol	Ordinal nominal	Drug prescribed and within-encounter dose status	No, Steady, Up, Down
troglitazone	Ordinal nominal	Drug prescribed and within-encounter dose status	No, Steady, Up, Down
tolazamide	Ordinal nominal	Drug prescribed and within-encounter dose status	No, Steady, Up, Down
examide	Ordinal nominal	Drug prescribed and within-encounter dose status	No, Steady, Up, Down
citoglipton	Ordinal nominal	Drug prescribed and within-encounter dose status	No, Steady, Up, Down
insulin	Ordinal nominal	Drug prescribed and within-encounter dose status	No, Steady, Up, Down
glyburide-metformin	Ordinal nominal	Drug prescribed and within-encounter dose status	No, Steady, Up, Down
glipizide-metformin	Ordinal nominal	Drug prescribed and within-encounter dose status	No, Steady, Up, Down
glimepiride-pioglitazone	Ordinal nominal	Drug prescribed and within-encounter dose status	No, Steady, Up, Down
metformin-rosiglitazone	Ordinal nominal	Drug prescribed and within-encounter dose status	No, Steady, Up, Down
metformin-pioglitazone	Ordinal nominal	Drug prescribed and within-encounter dose status	No, Steady, Up, Down
change	Binary nominal	Any diabetes medication change, including dose or generic name	Ch, No
diabetesMed	Binary nominal	Any diabetes medication prescribed	Yes, No
readmitted	Outcome	Time to recorded inpatient readmission	<30, >30, NO

Table 2. Data dictionary for the original variables in the Diabetes 130 US Hospitals dataset.

2.3. Initial data quality and cohort integrity checks

Before performing EDA, source-wide checks were performed. The raw file contained 101,766 rows, no fully duplicated rows, 101,766 unique encounter IDs, and no duplicated encounter IDs. However, only 71,518 unique patients were present, meaning 30,248 rows were additional encounters from repeat patients.

Check	Result	Implication
Rows / encounters	101,766	Raw encounter-level source size.
Fully duplicated rows	0	No exact duplicate records to remove.
Unique encounter IDs	101,766	Encounter identifier integrity passed.
Unique patients	71,518	Repeated longitudinal encounters exist.
Additional repeat encounters	30,248	Patient-level cohort definition required before splitting.

Table 3. Initial data-quality and cohort-integrity checks for the source dataset.

Missingness was encoded as `?` and converted to actual null values. In the raw categorical audit, weight was missing in 98,569 records (96.86%), medical specialty in 49,949 (49.08%), payer code in 40,256 (39.56%), race in 2,273 (2.23%), diag_3 in 1,423 (1.40%), diag_2 in 358 (0.35%), and diag_1 in 21 (0.02%).

For `A1Cresult` and `max\_glu\_serum`, the source value `None` did not mean that an observed laboratory result was accidentally lost. It explicitly means the test was not performed. Therefore this could be predictive on its own, and consequently `None` was recoded to `Not\_measured` and retained as a valid category. This preserves a clinically meaningful process signal: ordering or not ordering a test may itself reflect attention, severity, workflow, or treatment context. In the raw data, HbA1c was not measured in 84,748 encounters (83.28%) and maximum serum glucose was not measured in 96,420 (94.74%). These were not imputed as numeric lab results.

3. Cohort Definition and Domain-Derived Features

3.1. One first encounter per patient

Records were sorted by `encounter\_id`, and only the first encounter for each `patient\_nbr` was retained. This means that every row is now a different patient, making it aligned with the independence assumption of conventional classifiers. The trade-off is that later encounters and longitudinal trajectories are discarded, and this limitation is discussed at the end of the chapter.

Encounters ending in hospice or death were then excluded using discharge disposition IDs 11, 13, 14, 19, 20, and 21. Considering subsequent readmission for such patients is not meaningful. After first-encounter selection and these exclusions, the analytic cohort contained 69,973 rows. This differs by 11 from the 69,984 reported by Strack et al.; the discrepancy should be reconciled in future work by explicitly comparing the precise disposition mapping and preprocessing sequence.

3.2. Identifier and administrative feature removal

Both `encounter\_id` and `patient\_nbr` were removed after they had served cohort-integrity and deduplication purposes. They are identifiers, not generalizable clinical predictors, and retaining them would allow arbitrary record identity to enter the model. `payer\_code` was also removed.

3.3. Published ICD-9 Diagnosis Grouping

`diag\_1`, `diag\_2`, and `diag\_3` fields were mapped to clinically interpretable diagnosis groups using the published grouping in Strack et al. The same mapping was applied independently to all three diagnosis positions, after which the raw codes were dropped. This transformation reduces sparse one-hot levels and improves interpretation, but possibly sacrifices diagnosis-level specificity. This decision was nevertheless made in support of dimensionality control.

Group	ICD 9 definition used
Circulatory	390-459 or 785
Respiratory	460-519 or 786
Digestive	520-579 or 787
Diabetes	250.xx
Injury	800-999
Musculoskeletal	710-739
Genitourinary	580-629 or 788
Neoplasms	140-239
Other	V/E codes and all other valid or unparsable codes
Unknown	Missing diagnosis

Table 4. ICD-9 diagnosis categories used to group the primary and secondary diagnosis codes.

Major medical specialty groups : The 72 observed specialty labels were reduced to six levels. Cardiology, Family/GeneralPractice, InternalMedicine, and Surgery-General were retained as major groups; missing values became 'Unknown'; every other recorded specialty became 'Other'. This preserves the largest or most clinically interpretable groups while reducing sparse levels. Th

Two engineered medication summaries : Two columns were added before splitting. 'sumtotal_diabetes_meds' counts medication columns with Steady, Up, or Down status and therefore measures the number of active diabetes agents. 'num_med_changes' counts columns with Up or Down and measures how many individual drug doses changed. The latter adds information beyond the source 'change' flag, which also captures changes in generic name or other medication changes not represented as an Up/Down status.

Engineered feature	Definition	Purpose
sumtotal_diabetes_meds	Count of 23 drug fields in Steady, Up, or Down	Overall diabetes medication burden; later dropped as redundant.
num_med_changes	Count of 23 drug fields in Up or Down	Intensity of dose adjustment; finally retained.

Table 5. Definitions and intended purposes of the engineered diabetes-medication features.

In the training data, the resulting distribution was Unknown 48.18%, Other 20.39%, InternalMedicine 15.23%, Family/GeneralPractice 7.11%, Cardiology 5.92%, and Surgery-General 3.17%.

4. Training-test split

EDA was intentionally divided into two stages. Before the split (as detailed thus far), only source validation and cohort-definition operations were performed: review, duplicate and identifier checks, missing-value interpretation, one-patient cohort construction, death/hospice exclusion, identifier removal, published diagnosis grouping, specialty grouping, and domain feature creation. These operations define what an observation means, but do not use features by looking at the dataset as a whole and by looking at target relationships.

After the above were performed, the binary target was then created and a stratified 80/20 split was performed with ``random_state=42``. This produced 55,978 training observations and 13,995 test observations, with identical 8.97% positive prevalence. All distributional comparisons by outcome, rare-level thresholds, utilization binning, medication exposure thresholds, redundancy assessment, PCA fitting, and final feature decisions were based on ``X_train`` or a temporary ``train_df = X_train + y_train``. The untouched test-set received the transformations learned from training and was reserved for final model evaluation. This separation limited information leakage and rendered credibility to the reported test performance as an estimate of generalization.

4.1. Training-only EDA

4.1.1. Missingness and zero-variance predictors

The training partition confirmed that weight remained 96.07% missing and race 2.78% missing; other values were complete after explicit laboratory and specialty re-coding. Weight was dropped because of this extreme missingness - any imputation would be dominated by assumptions and a missingness indicator would contain far more information than actual weight. Race was retained for predictive evaluation and later fairness auditing; missing race was standardized to ``Unknown``.

Drug-level frequency checks showed that ``examide``, ``citoglipton``, and ``glimepiride-pioglitazone`` were 100% ``No`` in training. They were removed because zero-variance predictors cannot discriminate outcomes and only inflate the encoded design matrix. Other near-zero-variance drugs were not immediately discarded - their exposure prevalence and outcome rates were examined before applying a reproducible 1% exposure threshold.

4.1.2. Numeric Features

4.1.2.1. Outcome-stratified boxplots were used to examine central tendency, spread, skew, and extreme values within the data without assuming normality. Median and interquartile range were then calculated. Early-readmitted patients had a median stay of 4 days versus 3 days for those who were not, 46 versus 44 laboratory procedures, and 15 versus 14 medications.

4.1.2.2. Procedure counts, diagnosis counts, and active diabetes-medication counts had the same median and IQR across groups. The utilization counts and medication-change count had median 0 in both groups, illustrating why median-only summaries were insufficient for zero-inflated predictors.

Numeric feature	No early readmission Median, Q1-Q3	Early readmission median, Q1-Q3	Interpretation
time_in_hospital	3, 2-6	4, 2-7	Slightly longer stays among positives.
num_lab_procedures	44, 31-57	46, 34-59	Small upward shift among positives.
num_procedures	1, 0-2	1, 0-2	No unadjusted median difference.
num_medications	14, 10-20	15, 11-21	Small upward shift among positives.
number_outpatient	0, 0-0	0, 0-0	Zero-inflated; inspect nonzero status and tail.
number_emergency	0, 0-0	0, 0-0	Zero-inflated; inspect grouped tail.
number_inpatient	0, 0-0	0, 0-0	Zero-inflated but strong outcome gradient.
number_diagnoses	8, 6-9	8, 6-9	No median difference.
sumtotal_diabetes_meds	1, 1-2	1, 1-2	No median difference; redundancy examined.
num_med_changes	0, 0-0	0, 0-1	Medication adjustment more frequent in positives.

Table 6. Median and interquartile range of numeric predictors by 30-day readmission status in the training set.

4.1.2.3. Distribution plots and skewness : Distribution plots and skewness was calculated for numerical cols (Appendix A, Figure 2A). The values are shown below and figures show the skewness for the top three highly skewed variables (Figure 1):

number_emergency	16.899238
number_outpatient	9.858446
number_inpatient	5.659423
num_med_changes	1.585275
num_medications	1.434947
num_procedures	1.236455
time_in_hospital	1.173311
sumtotal_diabetes_meds	0.662592
num_lab_procedures	-0.215960
number_diagnoses	-0.720953

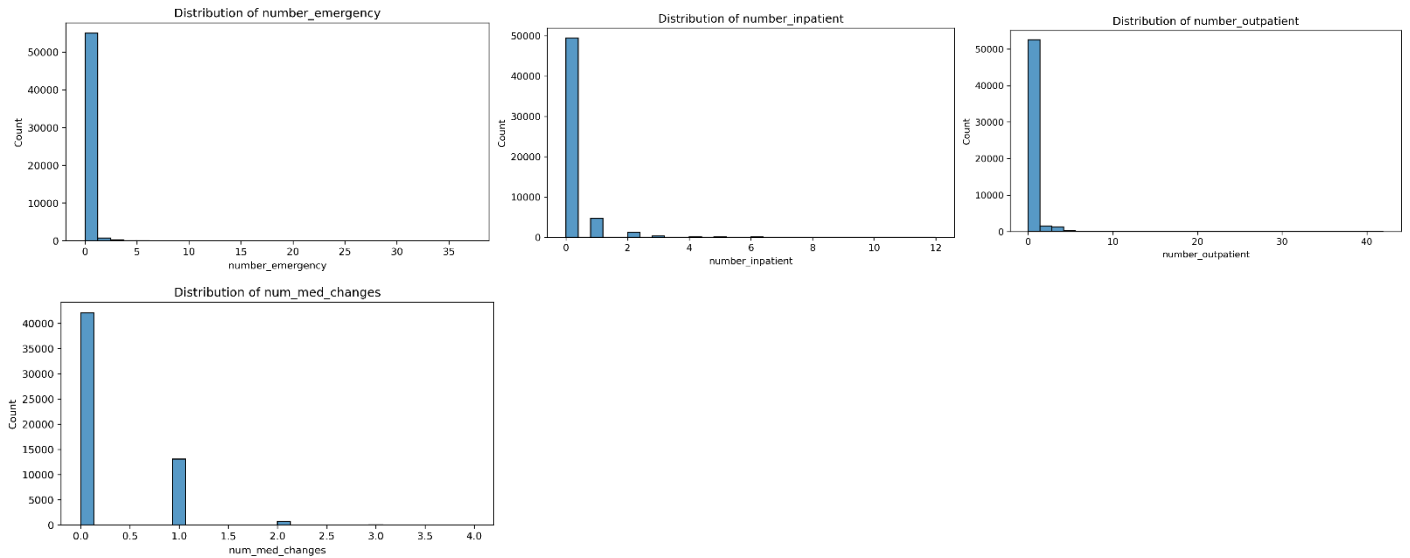


Figure 1. Training-set distributions of previous-year emergency, outpatient, and inpatient encounters, showing substantial right skew and concentration at zero.

The prior healthcare utilization variables exhibited substantial positive skew, particularly `number_emergency` (16.90), `number_outpatient` (9.86), and `number_inpatient` (5.66). This was primarily attributable to the high proportion of patients with no previous utilization (Table 6).

`'num_med_changes'` was highly concentrated at zero (Table 6, Figure 1): 75.17% of training observations had no Up/Down drug status, 23.40% had one, 1.32% had two, and only 0.10% had three. Nevertheless, the outcome cross-tab (Table 7) supported retaining it : one medication change occurred in 26.60% of positives versus 23.09% of negatives; two changes occurred in 1.49% versus 1.31%, and three in 0.18% versus 0.09%. This suggests medication modification may still contain predictive information, although the magnitude of the unadjusted difference is modest.

<code>'num_med_changes'</code>	Readmitted within 30 days	Not Readmitted within 30 days
0	75.51	71.72
1	23.09	26.6
2	1.31	1.49
3	0.09	0.18
4	0	0

Table 7. Distribution of the number of diabetes-medication dose changes by 30-day readmission status.

Because prior outpatient, emergency, and inpatient counts were dominated by zeros and were skewed, three additional features were temporarily created: binary `'any_outpatient'`, `'any_emergency'` and `'any_inpatient'` features. These were then cross-tabulated against the outcome. The goal was to determine whether the clinically meaningful distinction was simply **any** prior use, or whether the **number** of prior encounters retained a stable gradient (Table 8). Prior inpatient use showed the largest separation: 20.11% of positives versus 10.84% of negatives had at least one prior inpatient admission. Differences were smaller for emergency use (9.92% versus 7.02%) and outpatient use (14.72% versus 12.85%).

Feature	Positive group with “any” use	Negative group with “any” use	Readmission rate <u>no</u> use	Readmission rate “any” use
Outpatient	14.72%	12.85%	8.80%	10.14%
Emergency	9.92%	7.02%	8.72%	12.22%
Inpatient	20.11%	10.84%	8.11%	15.46%

Table 8. Previous-year healthcare utilisation and 30-day readmission rates among patients with and without prior use.

The binary versions were then compared with readmission rates at each original count (output shown on notebook). Outpatient counts showed no stable monotonic pattern and became extremely noisy above a few visits, so a binary `any_number_outpatient` feature replaced the count.

Emergency counts showed an increasing pattern over the well-populated early tail but very sparse high values; **counts were therefore grouped as 0, 1, 2, and 3 or more.**

Inpatient counts showed a clearer dose-response over 0, 1, 2, and 3 visits, followed by a sparse tail; **they were grouped as 0, 1, 2, 3, and 4 or more.** This preserved more clinically useful ordering than a binary inpatient indicator while limiting the influence of isolated extreme counts.

4.1.3. Categorical Feature EDA

Numeric ID fields for admission type, discharge disposition, and admission source were explicitly treated as categorical. For every categorical predictor, frequency, percentage, sample size, and unadjusted readmission rate by level were examined. A separate scan flagged levels below 1% so that striking rates from tiny groups would not be overinterpreted. Examples include discharge disposition IDs with only two or three observations, rare admission-source codes, ‘Unknown/Invalid’ gender with two training rows, the youngest age bands, and many medication dose-change levels.

Several clinically plausible patterns emerged: readmission increased with age, discharge disposition contained substantial rate variation, Family/GeneralPractice and InternalMedicine had higher unadjusted rates than Cardiology, and diagnosis groups differed across each diagnosis position. However, these are descriptive associations affected by case mix and sample size. Rare categories were not manually removed from all predictors. **For the three coded administrative fields, the final encoder grouped levels occurring below 1% through `OneHotEncoder(handle_unknown="infrequent_if_exist", min_frequency=0.01)`. Other categorical fields used one-hot encoding with unknown levels ignored; explicit ‘Unknown’ categories preserved missing sensitive-group and specialty information.**

4.1.3.1. Medication EDA : Medication EDA combined prevalence and outcome rate (results printed in notebook). Insulin had substantial exposure and a consistent descriptive gradient: early-readmission rates were 8.28% for ‘No’, 9.21% for ‘Steady’, 10.01% for ‘Up’, and 10.53% for ‘Down’. Metformin, glipizide, glimepiride, repaglinide, pioglitazone, and rosiglitazone also showed differences, but many Up/Down cells were small. Extremely high or zero rates for

drugs such as acarbose or combination products were driven by single-digit denominators and were not treated as reliable evidence (cell 30, notebook).

The final representation combined 'Up' and 'Down' as 'Dose_changed', reducing direction-specific sparsity while retaining 'No' and 'Steady'. Exposure was defined as any value other than 'No' and measured on training data only. After the three zero-variance drugs had already been dropped, a 1% minimum exposure threshold removed 12 additional medication columns: acetohexamide, metformin-pioglitazone, metformin-rosiglitazone, troglitazone, glipizide-metformin, tolbutamide, miglitol, tolazamide, chlorpropamide, acarbose, nateglinide, and glyburide-metformin. Their training exposure ranged from 0.0018% to 0.7074%. The retained individual drug predictors were metformin, repaglinide, glimepiride, glipizide, glyburide, pioglitazone, rosiglitazone, and insulin.

4.1.3.2. Correlation Matrix and VIF : A Spearman correlation matrix was used for numeric predictors because several distributions were non-normal, skewed, or ordinal-like (Figure 2).



Figure 2. Spearman correlation matrix for the numeric candidate predictors in the training set.

VIF was also calculated as an exploratory linear-collinearity diagnostic after median filling. All numeric VIFs were low: 'num_medications' was highest at 1.67, followed by 'time_in_hospital' at 1.39; every other VIF was 1.22 or lower. These results together did not support excluding a numeric variable solely for multicollinearity.

Numeric predictor	VIF
num_medications	1.67
time_in_hospital	1.39
num_procedures	1.22
sumtotal_diabetes_meds	1.16
num_lab_procedures	1.16
num_med_changes	1.16
number_diagnoses	1.11
number_inpatient	1.03
number_emergency	1.03
number_outpatient	1.02

Table 9. Variance inflation factors for the numeric candidate predictors.

The relationship between `change` and `num\_med\_changes` received a separate audit (Table 10). Every nonzero medication-change count implied `change = Ch`, but 11,365 training encounters had `change = Ch` while `num\_med\_changes = 0`. This occurs because the source flag captures medication changes beyond Up/Down statuses. Their Spearman correlation was 0.63 and both VIFs were 1.59, so they were related but not redundant. Both were retained.

change	num_med_changes					
	0	1	2	3	4	All
Ch	11365	13100	740	57	2	25264
No	30714	0	0	0	0	30714
All	42079	13100	740	57	2	55978

Table 10. Cross-tabulation of the source medication-change indicator and the engineered medication-change count.

4.1.3.3. Diabetes medication indicator vs active medication count :

`diabetesMed` and `sumtotal\_diabetes\_meds` were deterministically related (Table 11): all 13,311 patients with `diabetesMed = No` had zero active diabetes medications, whereas all 42,667 with `Yes` had one to six. Readmission was 7.58% at zero drugs and roughly 8.66%-9.59% for one to four drugs. Counts of five and six were only 37 and four, yielding unstable rates of 0% and 50%. The main stable distinction was therefore none versus any treatment. `diabetesMed` was retained and `sumtotal\_diabetes\_meds` was dropped to reduce redundancy and avoid sparse-tail noise.

diabetesMed	sumtotal_diabetes_meds							
	0	1	2	3	4	5	6	All
No	13311	0	0	0	0	0	0	13311
Yes	0	24948	12300	4581	797	37	4	42667
All	13311	24948	12300	4581	797	37	4	55978

Table 11. Relationship between diabetes-medication status and the number of active diabetes medications.

5. EDA-Informed Final Feature Engineering

5.1. Final Decision - Summary

Decision	Evidence and rationale
Drop weight	96.07% missing in training; observed values too sparse for credible imputation.
Drop examide, citoglipton, glimepiride-pioglitazone	Zero variance in training.
Drop sumtotal_diabetes_meds	Deterministic relationship with diabetesMed and little stable gradient among 1-4 drugs.
Replace number_outpatient	Binary 'any' use was more stable than a noisy nonmonotonic sparse tail.
Group number_emergency	Categories 0, 1, 2, 3+ preserve early-tail gradient and limit extreme sparsity.
Group number_inpatient	Categories 0, 1, 2, 3, 4+ preserve the strongest utilization gradient.
Standardize race and gender absence	Missing race and Unknown/Invalid gender become explicit Unknown for fairness and modeling.
Combine drug Up and Down	Dose_changed reduces rare direction-specific levels.
Drop drugs below 1% exposure	Training-only prevalence rule removes unreliable sparse medication features.
Retain change and num_med_changes	They capture overlapping but distinct medication-change information; VIF 1.59.
Retain A1Cresult and max_glu_serum Not_measured	Absence means that test not performed, and may be informative.
Retain medical_specialty Unknown	Missingness may reflect workflow; major-group reduction controls cardinality.

Table 12. Summary of EDA-informed feature-engineering and feature-selection decisions.

5.2. Final Feature Set and preprocessing

After EDA-informed transformations, the same 31 predictors were now in both train and test sets, and in the same order. There were 24 categorical and seven numeric predictors. Assertions were run to ensure feature alignment, row-count and index alignment, binary nonmissing targets, absence of duplicate feature names, successful removal of all excluded variables, presence of engineered features, standardized sensitive-group labels, no remaining Up/Down labels in retained drug columns, and disjoint train/test indices. All checks passed.

Type of feature	Final predictors
Numeric (7)	time_in_hospital; num_lab_procedures; num_procedures; num_medications; number_diagnoses; num_med_changes; any_number_outpatient
Categorical (24)	race; gender; age; admission_type_id; discharge_disposition_id; admission_source_id; medical_specialty; max_glu_serum; A1Cresult; metformin; repaglinide; glimepiride; glipizide; glyburide; pioglitazone; rosiglitazone; insulin; change; diabetesMed; diag_1_group; diag_2_group; diag_3_group; number_emergency_cat; number_inpatient_cat

Table 13. Final numeric and categorical predictors included in the modelling pipeline.

The reproducible preprocessing pipeline median-imputes and standardizes numeric predictors.

It constant-imputes other categorical predictors with 'Unknown' and one-hot encodes them with unseen levels ignored. Admission type, discharge disposition, and admission source use most-frequent imputation and infrequent-category grouping at a 1% training frequency. Because preprocessing is inside the later model pipeline, parameters are learned within training folds during model development rather than from the test data.

5.3. Feature selection and dimensionality reduction

Thus far, domain and filter rules removed identifiers, extreme-missingness fields, zero-variance variables, rare medication exposures, unstable tails, and a deterministically redundant engineered count. The subsequent model-development workflow uses embedded coefficient-magnitude selection within logistic regression; this is described with model implementation rather than claimed as pre-model EDA.

PCA was fitted to the seven standardized final numeric predictors as an exploratory dimensionality-reduction assessment. PC1 loaded most strongly (see below) on number of medications (0.556), time in hospital (0.512), laboratory procedures (0.373), procedures (0.334), and number of diagnoses (0.332), suggesting a broad encounter-intensity axis. PC2 was dominated by any prior outpatient use (0.876) (output on notebook). The first two components did not visibly separate readmission classes (see appendix A, Figure 3A), and PCA would not cover the many clinically meaningful categorical variables without changing the representation. PCA was therefore documented but not used to replace the original features, preserving interpretability and nonlinear modeling flexibility.

Loadings for PC1

num_medications	0.556353
time_in_hospital	0.512022
num_lab_procedures	0.372863
num_procedures	0.334460
number_diagnoses	0.331926
num_med_changes	0.258752
any_number_outpatient	0.016936

6. Limitations/important caveats of data and feature selection strategies

- Readmissions outside the participating health system may be missed. The 'NO' outcome therefore means no recorded readmission, not necessarily no readmission anywhere.
- Keeping only the first encounter improves independence and prevents leakage, but discards longitudinal information and changes the task from encounter-level risk to first-eligible-encounter risk. A patient-grouped split with longitudinal features may be a useful alternative design.
- The final cohort count is 11 lower than the source paper's 69,984. For future work, the discharge-disposition mapping and sequence of exclusions should be reconciled.
- The 1% drug-exposure threshold and utilization bins are pragmatic and training-derived, but not tuned through nested validation. Sensitivity analyses could compare alternative thresholds and data-driven binning.

- Collapsing “Up” and “Down” improves stability but removes potentially meaningful dose direction. A hierarchical or regularized model could retain direction while shrinking rare effects.

7. Model Implementation

7.1. General Validation Design and Final Model Selection Strategy

For this project, three models were compared : logistic regression, random forest and XGBoost. Model development used only the 55,978 observations of the training partition. For each model, five-fold stratified cross-validation was performed which divided the training data into five approximately equal subsets while preserving the 8.97% positive-class prevalence in each fold. At each iteration, four folds trained the complete pipeline and the remaining fold evaluated it. This means that every fold served once as validation data. Averaging across folds gives a more stable estimate than a single validation split and exposes each model to several train-validation combinations. The common five-fold object used `'StratifiedKFold(n_splits=5, shuffle=True, random_state=42)'`. Pipelines kept preprocessing inside the validation procedure so that imputation, scaling, one-hot encoding, infrequent-level handling, and any feature selection were learned from the training portion of each fold. Average Precision was the refit and model-selection metric because the positive class was uncommon. ROC-AUC and Brier loss assessed ranking and probability accuracy; recall, precision, specificity, F1, and balanced accuracy assessed decisions at a specified threshold.

To trial different combinations of hyperparameters, GridSearchCV was used for hyperparameter tuning for the Logistic Regression model, and RandomizedSearchCV was used for Random Forest and XGBoost

Upon finding the best hyperparameters, Out-of-fold probabilities are calculated. Out-of-fold, or OOF, probabilities are probabilities for every training observation from a model that did not train on that observation. `'cross_val_predict'` generated these probabilities for each person by using a model that did not train on that person. Average Precision, ROC-AUC, and Brier loss were calculated directly from the model's pooled out-of-fold predicted probabilities and are independent of any single classification threshold.

Obtaining OOF probabilities then in turn supported threshold selection (which in turn was decided by Youden's J). Recall, precision, specificity, F1 score, and balanced accuracy were calculated after converting those pooled probabilities into class predictions using each model's independently selected Youden-optimal threshold.

It must be noted that this strategy means that Hyperparameters were first selected using cross-validation across the **complete training set**, after which the chosen configurations were used to generate post-tuning OOF predictions. Because model and hyperparameter selection had already used information from the same overall cross-validation folds, these post-tuning OOF estimates may be mildly optimistic. The untouched test set therefore provides the more rigorous final estimate of the selected model's performance. This is further discussed in section 7.7.

The 13,995-observation test-set remained untouched throughout feature decisions, tuning, model selection, and threshold optimization. All 3 models' metrics were then compared, and only when the final model had been selected, was the fitted winning pipeline applied to the test set once.

7.2. Dummy Classifier Benchmark

A `DummyClassifier(strategy="prior")` was first used to establish a “no-skill baseline” that only knows the class distribution. The “prior” method learned that about 9% of training patients were readmitted, so it assigned a predicted probability of ~0.09 to every single patient (Table 14). Since 0.09 is well below the

default 0.50 decision threshold that sklearn uses to convert probabilities into class labels, every prediction comes back as "no readmission." Its purpose is to establish the performance available without predictive information. A real model should materially improve on this benchmark, especially the prevalence-level Average Precision and chance-level ROC-AUC.

Metric	Dummy classifier CV result
Average Precision	0.090
ROC-AUC	0.500
Recall at 0.50	0.000
Precision at 0.50	0.000
Specificity at 0.50	1.000
F1 at 0.50	0.000
Balanced accuracy	0.500
Brier loss	0.082

Table 14. Cross-validated performance of the dummy classifier.

7.3. Logistic Regression Development

After the dummy classifier was established as the “no-skill baseline”, logistic regression was considered next. Logistic regression models the log-odds of early readmission as a linear combination of encoded predictors and converts the result to a probability. It was chosen as the interpretable baseline because its coefficients have directional and odds-ratio interpretations, regularization can control a large one-hot-encoded feature space, and it provides a strong check on whether more flexible tree models add useful nonlinear structure. It also establishes a transparent reference for later explainability and deployment discussions.

7.3.1. Unweighted baseline

The first genuine model used the complete preprocessing pipeline with `class_weight=None`, a maximum of 3,000 iterations, and the default 0.50 threshold. `cross_validate` was used to answer the question “which class-weighting option has better average validation performance across the five folds?”.

Five-fold CV produced Average Precision 0.161 and ROC-AUC 0.644 (Table 15), a substantial ranking improvement over the dummy model (these metrics are the average metric of 5 folds). Brier loss improved slightly from 0.082 to 0.080. However, recall was only 0.001, F1 was 0.002, and balanced accuracy remained 0.500. This does not mean the probabilities contained no signal; it means that with a 9% positive class, almost no predicted probability exceeded 0.50. The result motivated separating probability ranking from the operating threshold.

7.3.2. Balanced versus unbalanced class weights

Metric	Unweighted	Balanced
Average Precision	0.161	0.160
ROC-AUC	0.644	0.645
Recall at 0.50	0.001	0.534
Precision at 0.50	0.207	0.138
F1 at 0.50	0.002	0.219
Balanced accuracy	0.500	0.602
Brier loss	0.080	0.230

Table 15. Cross-validated performance of unweighted and class-balanced logistic regression at the 0.50 threshold.

Balanced class weights upweighted positive observations inversely to their frequency. At the default threshold, this moved many probabilities above 0.50 and increased recall from 0.001 to 0.534, but precision fell to 0.138 and Brier loss deteriorated sharply to 0.230. Ranking barely changed: Average Precision moved from 0.161 to 0.160 and ROC-AUC from 0.644 to 0.645.

Conclusion : Class weighting therefore changed probability scale and classification behavior more than discrimination.

7.3.3 L1 feature selection versus no selection

L1 regularization adds a penalty proportional to the sum of absolute coefficient magnitudes. It can shrink weak coefficients exactly to zero, which is useful after one-hot encoding creates many correlated or low-information levels. Two pipelines were made : one without an L1 estimator, and one with an L1 logistic estimator, which was used inside 'SelectFromModel'; selection was based on coefficient magnitude relative to the selector threshold, not simply on whether a coefficient was nonzero. In this case, the selected subset was then passed to a separate final logistic regression. 'cross_val_predict' was used here to see how two pipelines behaved when each patient received an out-of-fold probability, and after applying a particular threshold. In retrospect, 'cross_validate' could have also been used here at this point.

Metric at 0.50	No feature selection	L1 feature selection
Average Precision	0.160	0.160
ROC-AUC	0.644	0.644

Metric at 0.50	No feature selection	L1 feature selection
Recall	0.001	0.001
Precision	0.222	0.389
Specificity	1.000	1.000
F1	0.002	0.003
Balanced accuracy	0.500	0.501
Brier loss	0.080	0.080

Table 16. Out-of-fold logistic-regression performance with and without L1-based feature selection at the 0.50 threshold.

OOF comparisons showed essentially identical ranking and probability performance. L1 selection improved the precision of the very few default-threshold positives, but default-threshold recall remained unusably low.

Conclusion : This suggested L1 feature selection might simplify the model without sacrificing predictive performance, so feature selection remained a hyperparameter rather than being either mandated or discarded.

7.3.4 Hyperparameter search and selected specification

The above two preliminary diagnostic experiments were conducted first to understand how each modeling choice affected behavior. A GridSearchCV was then consequently performed to evaluate 96 logistic configurations over five folds, for 480 fits.

The search compared :

1. No separate feature-selection step versus L1 'SelectFromModel'; final-model penalties L1 versus L2; regularization strengths C of 0.01, 0.1, 1, and 10;
2. Class weights None versus balanced; and selector C values from 0.01 to 1 (smaller C implies stronger regularization). The search refit the configuration with the highest mean CV Average Precision.

The winning specification used:

L1 feature selection with selector C = 0.1 (controls the L1 log regression inside SelectFromModel, its purpose is to decide which encoded features are retained)

L1 logistic regression with C = 1 (a separate L1 log regression with C=1 was fitted using those retained features)

Class weighting = None

The selector reduced 129 encoded columns to 72 and removed 57. Its mean CV Average Precision was 0.16158.

The top ten configurations were tightly clustered in Average Precision, approximately 0.16114 - 0.16158, so the numerical advantage of the winner was small. Their default-threshold recalls were also near zero; this reinforced the need for OOF threshold optimization rather than suggesting that hyperparameter tuning had solved the operating-point problem.

The tuned logistic pipeline was then used to generate OOF probabilities (using `'cross_val_predict'`) for every training observation. The ROC curve plots sensitivity against false-positive rate across thresholds, while the precision-recall curve focuses on the minority positive class and displays the trade-off between identifying more readmissions and preserving positive predictive value. Their summary areas were ROC-AUC 0.645 and Average Precision 0.160: better than no-skill, but only moderate discrimination. Figure 3 shows the ROC and PR curves.

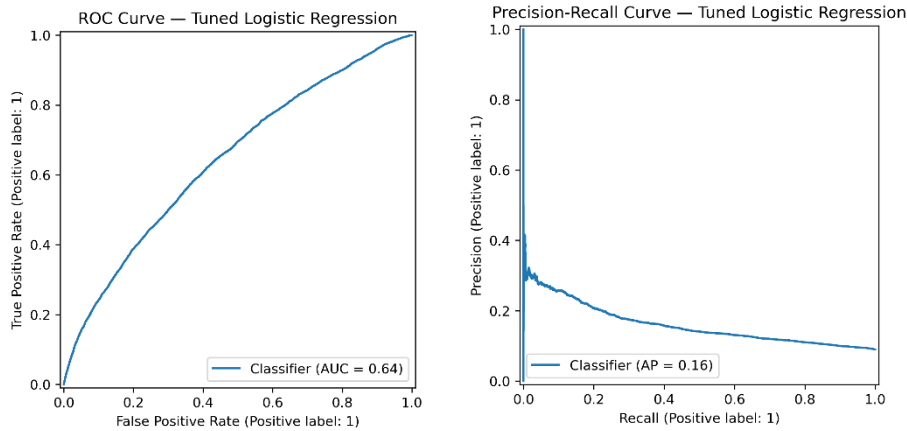


Figure 3. Out-of-fold ROC and precision-recall curves for the tuned logistic-regression model, including the selected operating threshold and no-skill reference.

The operating threshold was then selected by maximizing Youden's J, defined as sensitivity plus specificity minus one, on OOF predictions.

The selected threshold was 0.0836

OOF recall 0.588,

specificity 0.622

J = 0.211.

Applying this threshold to create predictions produced :

Metric	Final Tuned Log Regression results using pooled OOF probabilities
Average Precision	0.160
ROC-AUC	0.645
Recall at 0.0836	0.588
Precision at 0.0836	0.133
Specificity at 0.0836	0.622

Metric	Final Tuned Log Regression results using pooled OOF probabilities
F1 at 0.0836	0.217
Balanced accuracy at 0.0836	0.605
Brier loss	0.08

Table 17. Out-of-fold performance of the tuned logistic-regression model at the selected threshold of 0.0836.

The OOF confusion matrix (Figure 4) therefore showed the intended change from the 0.50 threshold: substantially more true positives, accompanied by many false positives because the positive class is rare.

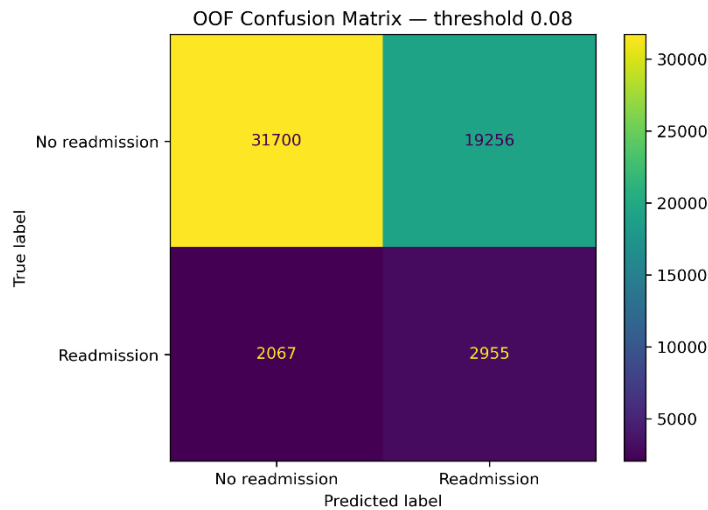


Figure 4. Out-of-fold confusion matrix for the tuned logistic-regression model at the selected threshold of 0.0836.

7.4. Random Forest Development

A random forest combines the results of many decision trees to make a prediction. Each tree is trained using a slightly different random sample of the data – this is called a bootstrap sample. Each tree also considers a random selection of predictors. This means that individual trees are able to capture nonlinear relationships and interactions without needing to pre-specify them/The random forest then averages the predictions from all the trees, which also means that it reduces the variance of a single deep tree. Random Forest was chosen to test whether readmission risk depended on nonlinear thresholds and interactions that logistic regression could not represent, while remaining a robust and widely used tabular-data benchmark.

7.4.1. Tuning Results and overfitting signal

RandomizedSearchCV tested 20 sampled configurations over five folds, for 100 fits. The search varied: Number of trees (200, 400, 600) , maximum depth (None, 10, 20, 30), minimum samples per split (2, 5, 10) and leaf (1,5, 10), feature subsampling (sqrt, Log2), and class weighting (None, balanced, balanced_subsample).

The winning forest used :

400 trees,
unlimited depth,
minimum split size 2,
minimum leaf size 5,
square-root feature sampling
no class weighting.

For each Random Forest hyperparameter configuration, the model was evaluated on both :

1. The data used to fit the model within each fold
2. The validation fold that the model did not use for fitting

This time the model was additionally checked between the data it trained on and the validation data because Random Forest has a greater capacity to memorize the training data than regularized logistic regression. A deep forest can create highly specific decision rules and fit noise, especially when:

- max_depth=None
- Trees can continue splitting until other stopping rules apply
- The dataset has many one-hot-encoded predictors
- Hundreds of trees are fitted
- The positive class is relatively rare

The metrics of the top-ranked RF configuration is shown in Table 19.

Metric	
Mean CV Average Precision	0.174687
Std CV Average Precision	0.001018
Mean train Average Precision	0.884105
Std train Average Precision	0.005592

Table 18. Cross-validation and training Average Precision of the selected random-forest configuration. “Mean CV Average Precision : the mean AP calculated from the validation folds (data from 1 validation fold, 5 iterations, and the mean of the 5 iterations), Std CV AP: Standard Deviation of AP of the validation folds, Mean train Average Precision : the mean AP calculated from the training folds (data from 4 training folds, 5 iterations, and the mean of the 5 iterations), Std train Average Precision: Standard Deviation of AP of the training folds

The winning configuration CV Average Precision was 0.17469, higher than logistic regression.

The top-ranked configuration had mean training Average Precision 0.884 versus Mean Validation 0.175, a large gap indicating **substantial overfitting** despite aggregation and a minimum leaf size of five. Other top configurations had lower training AP, from 0.287 to 0.600, with similar validation AP (output on notebook). This gap is a reason to interpret the small performance advantage cautiously and to consider stronger depth or leaf regularization in future nested tuning, if this was the winning model.

7.4.1. OOF threshold curves and Confusion Matrix

OOF probabilities were generated from the tuned forest using `'cross_val_predict'`, produced Average Precision 0.173, ROC-AUC 0.647, and Brier loss 0.079 (Table 19). The Youden-optimal threshold was 0.0903 and Youden's $J = 0.210$. At this threshold, recall, specificity, precision, F1 and balanced accuracy are listed in Table 19.

Metric	Final Tuned Random Forest results using pooled OOF probabilities
Average Precision	0.173
ROC-AUC	0.647
Recall at 0.0903	0.610
Precision at 0.0903	0.131
Specificity at 0.0903	0.600
F1 at 0.0903	0.215
Balanced accuracy at 0.0903	0.605
Brier loss	0.079

Table 19. Out-of-fold performance of the tuned random-forest model at the selected threshold of 0.0903.

Relative to logistic regression, the forest ranked positives somewhat better and recovered slightly more readmissions, but it also classified more negatives as positive.

7.5. XGBoost development

7.5.1 Model description and rationale

XGBoost is gradient-boosted decision-tree modeling. Trees are added sequentially so that each new tree corrects residual errors left by the current ensemble. Learning rate, tree depth, row subsampling, and column subsampling regularize the process. XGBoost was chosen because boosted trees are strong performers on heterogeneous tabular data, can learn nonlinearities and high-order interactions, and often improve minority-class ranking without requiring manual interaction terms.

7.5.2 Hyperparameter search results

RandomizedSearchCV evaluated 20 sampled configurations over five folds, again totaling 100 fits and optimizing Average Precision. The search varied: Number of trees (200, 400, 600), maximum depth (3, 5, 7), learning rate (0.01, 0.05, 0.10), row subsampling (0.7, 0.9, 1.0), and column subsampling (0.7, 0.9, 1.0).

The winning configuration used :

600 trees
maximum depth 7
learning rate 0.01
subsample 0.70
column subsample 0.70.

The metrics of the top ranked configuration is shown below :

Metric	
Mean CV Average Precision	0.180826
Std CV Average Precision	0.002775
Mean train Average Precision	0.429379
Std train Average Precision	0.004451

Table 20. Cross-validation and training Average Precision of the selected XGBoost configuration. “Mean CV Average Precision : the mean AP calculated from the validation folds (data from 1 validation fold, 5 iterations, and the mean of the 5 iterations), Std CV AP: Standard Deviation of AP of the validation folds, Mean train Average Precision : the mean AP calculated from the training folds (data from 4 training folds, 5 iterations, and the mean of the 5 iterations), Std train Average Precision: Standard Deviation of AP of the training folds.

The validation AP values of the five leading configurations ranged from 0.17573 to 0.18083 (output shown on notebook). The winning configuration’s training AP was 0.429 versus validation AP 0.181 (Table 20), again showing a train-validation gap, although much smaller than the top random forest gap.

7.5.3 OOF threshold curves and confusion matrix

`Cross\_val\_predict` generated OOF probabilities using the tuned XGBoost model. AP, ROC-AUC and Brier loss were calculated (Table 21). The resulting ROC Curve was then used to again maximize Youden’s J (0.22), and the selected threshold was 0.0788. Recall, Precision, Specificity, F1 and Balanced Accuracy was then calculated using this threshold (Table 21).

Metric	Final Tuned XGBoost results using pooled OOF probabilities
Average Precision	0.180
ROC-AUC	0.654
Recall at 0.0788	0.667
Precision at 0.0788	0.128
Specificity at 0.0788	0.552
F1 at 0.0788	0.215
Balanced accuracy at 0.0788	0.610
Brier loss	0.079

Table 21. Out-of-fold performance of the tuned XGBoost model at the selected threshold of 0.0788.

The tuned XGBoost OOF probabilities achieved Average Precision 0.180, ROC-AUC 0.654, and Brier loss 0.079 (Table 21).

The ROC curve therefore showed the strongest overall ranking of the compared models, and the precision-recall curve showed the largest improvement over the 0.09 prevalence baseline.

7.6. Training set model comparison and selection

Logistic Regression, Random Forest and XGBoost were evaluated by combining all OOF probabilities, applying a classification threshold and calculating metrics once over the pooled predictions. The metrics in the original Dummy summary (Table 14) was calculated separately in each validation fold and then averaged. Therefore, the dummy model was recalculated using `cross\_val\_predict` similarly to the other 3 models. The table below show the adjusted Dummy metrics, and makes a comparison of all 4 models.

Metric	Dummy	Logistic regression	Random forest	XGBoost
Average Precision	0.090	0.160	0.173	0.180
ROC-AUC	0.500	0.645	0.647	0.654
Recall	0.000	0.588	0.610	0.667
Precision	0.000	0.133	0.131	0.128
Specificity	1.000	0.622	0.600	0.552
F1	0.000	0.217	0.215	0.215
Balanced accuracy	0.500	0.605	0.605	0.610
Brier loss	0.082	0.080	0.079	0.079

Table 22. Comparison of pooled out-of-fold performance across the dummy, logistic-regression, random-forest, and XGBoost models.

XGBoost was then selected as the final model, because it had the highest primary metric (Average Precision), training CV Average Precision, as well as the highest ROC-AUC, recall, balanced accuracy, and Youden J at its independently selected threshold. Its AP improvement over random forest was modest, and all substantive models had low precision at sensitivity-oriented thresholds. Therefore, this is the best candidate among those tested, not the most clinically definitive model. Logistic regression remained the most interpretable benchmark; random forest provided intermediate AP but showed a particularly large train-validation gap.

7.7. Final evaluation on the untouched test set

The selected XGBoost pipeline was retained with its OOF-selected threshold of 0.0788. It produced probabilities for the 13,995 untouched test observations, and the fixed threshold converted those probabilities to classifications. Neither the model family, hyperparameters, preprocessing decisions, nor threshold were changed after seeing test performance.

Metric	XGBoost OOF validation	Final held-out test	Test minus OOF
Average Precision	0.180	0.183	+0.003
ROC-AUC	0.654	0.653	-0.001
Recall	0.667	0.666	-0.001
Precision	0.128	0.128	0.000
Specificity	0.552	0.555	+0.002
F1	0.215	0.215	0.000
Balanced accuracy	0.610	0.610	0.000
Brier loss	0.079	0.079	approximately 0.000

Table 23. Comparison of XGBoost performance on the out-of-fold validation predictions and the untouched test set.

The final test confusion matrix contained 7,065 true negatives, 5,675 false positives, 419 false negatives, and 836 true positives.

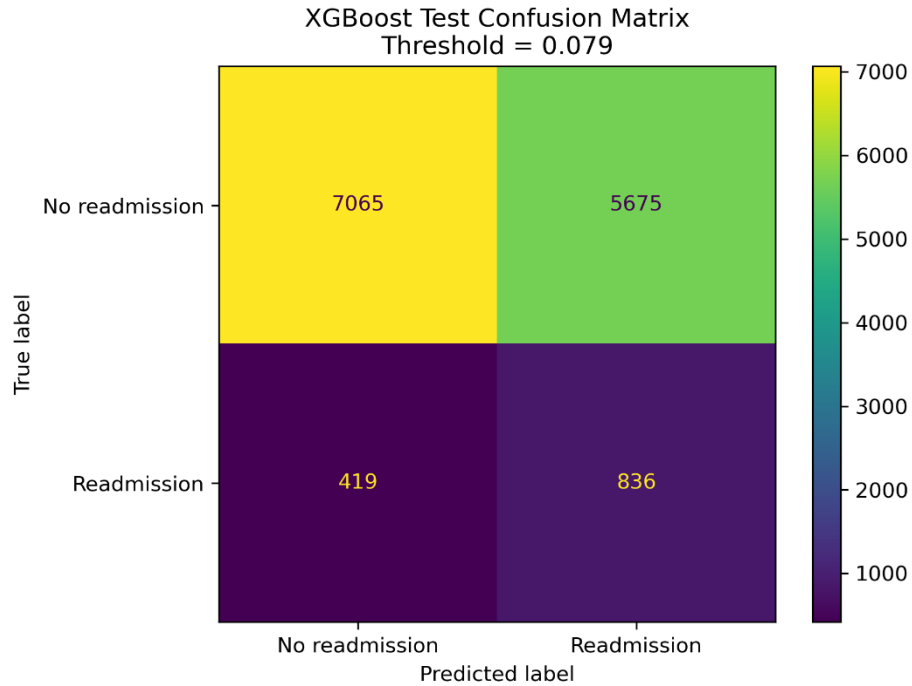


Figure 5. Confusion matrix for the final XGBoost model on the untouched test set at the fixed threshold of 0.0788.

Thus, the model identified 836 of 1,255 early readmissions, giving 66.6% recall, while its 6,511 positive flags contained 12.8% true readmissions. Specificity was 55.5%. The ROC-AUC of 0.653 indicates moderate ranking discrimination, and the test Average Precision of 0.183 was 2.04 times the 0.090 prevalence baseline.

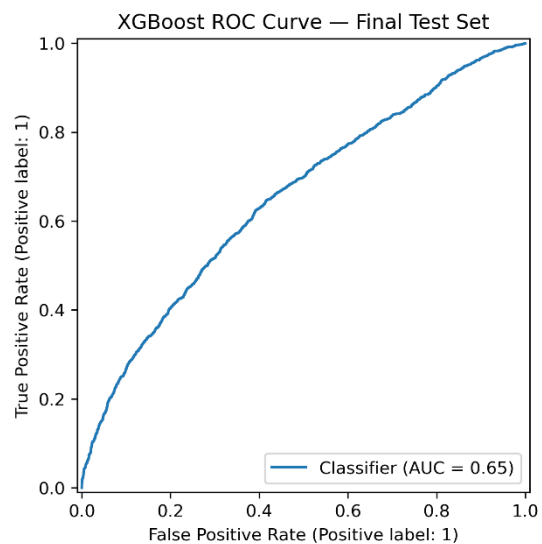
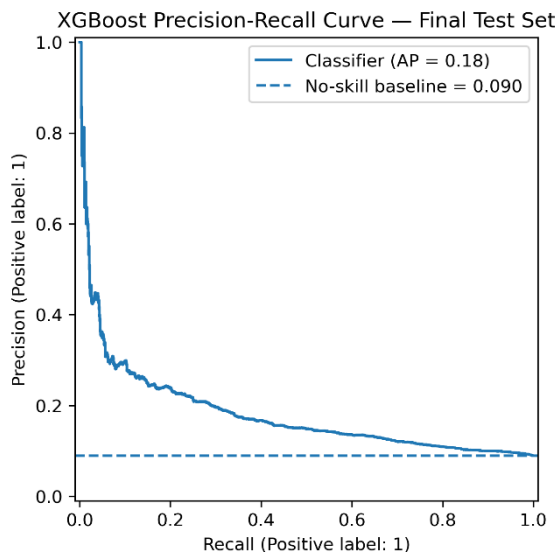


Figure 6. ROC and precision-recall curves for the final XGBoost model on the untouched test set, with the no-skill prevalence reference shown on the precision-recall curve.

The precision-recall curve is especially important here (Figure 6) because it shows how precision deteriorates as recall increases in the imbalanced setting; the dashed 0.090 line represents no-skill precision.

Test and OOF metrics were nearly identical across every measure. Test AP was only 0.003 higher, ROC-AUC and recall were 0.001 lower, specificity was 0.002 higher, and F1, balanced accuracy, precision, and Brier loss were effectively unchanged. This close agreement supports stable generalization to the held-out sample. It does not establish transportability to new hospitals, later years, or different care systems.

The lower threshold prioritized sensitivity: about two-thirds of early readmissions were identified, but approximately 45% of negatives were also flagged. The corresponding OOF confusion matrix would therefore contain more true positives and false positives than the logistic or random-forest matrices. This operating point can be reasonable for a low-risk supportive intervention, but not necessarily for a costly or burdensome intervention.

7.7. Discussion of Limitations in model development

1. Training versus validation performance could have been checked for Logistic Regression as well, not just Random Forest and XGBoost
2. Possibly optimistic OOF probabilities were briefly mentioned in section 7.1 due to selection of hyperparameters using cross-validation across the complete training set. This could have been improved by using nested cross-validation :
 - a. Outer CV estimates performance.
 - b. Inner CV selects hyperparameters.
 - c. Each validation fold remains completely uninvolved in model and hyperparameter selection.

Nevertheless, it was observed that Test-set and OOF metrics on the validation set were nearly identical, so the approach used in this project was still effective.

3. Youden's J is a neutral statistical criterion; actual deployment should also incorporate intervention capacity and the relative costs of missed and unnecessary interventions.

8. SHAP AND PDP EXPLAINABILITY

The final XGBoost model was interpreted with SHAP (Shapley Additive exPlanations) and Partial Dependence. These methods answer complementary questions. SHAP decomposes an individual model output into feature contributions and can be aggregated to describe global model behavior. Partial dependence on the other hand estimates the model's average predicted response as one feature is varied across the observed sample.

8.1. SHAP Method

The workflow essentially did three things:

1. **Explain the predictions** — identify which features push predicted risk higher or lower.
2. **Validate the explanations** — confirm that the SHAP calculations correspond exactly to the fitted pipeline and XGBoost predictions.
3. **Understand the model globally** — determine which features the model relies on most across the test population.

The fitted preprocessing and XGBoost pipeline was first preserved. A reproducible random sample of 1,000 untouched test observations was transformed using the training-fitted preprocessor, producing 129 encoded features. TreeExplainer then calculated SHAP values for the fitted XGBoost estimator. The implementation verified that the transformed representation reproduced the pipeline probabilities and that the SHAP base value plus all feature contributions reconstructed each XGBoost raw margin to within 4.77×10^{-6} (because SHAP baseline + all the patient's SHAP values $\approx$ model output). SHAP values are reported on the raw margin, or log-odds, scale: positive values push a prediction toward higher risk and negative values push it toward lower risk. Their magnitudes are not percentage-point changes in probability.

Global importance was measured as mean absolute SHAP value across the 1,000-patient sample. This describes the average magnitude of a feature level's contribution but not its direction. The beeswarm plot adds direction and shows the distribution of contributions. Because categorical predictors were one-hot encoded, each category level appears separately and the importance of an original variable may be spread across several levels. Related predictors can also share importance; rankings should therefore be treated as descriptions of this fitted model, not as a definitive hierarchy of clinical causes.

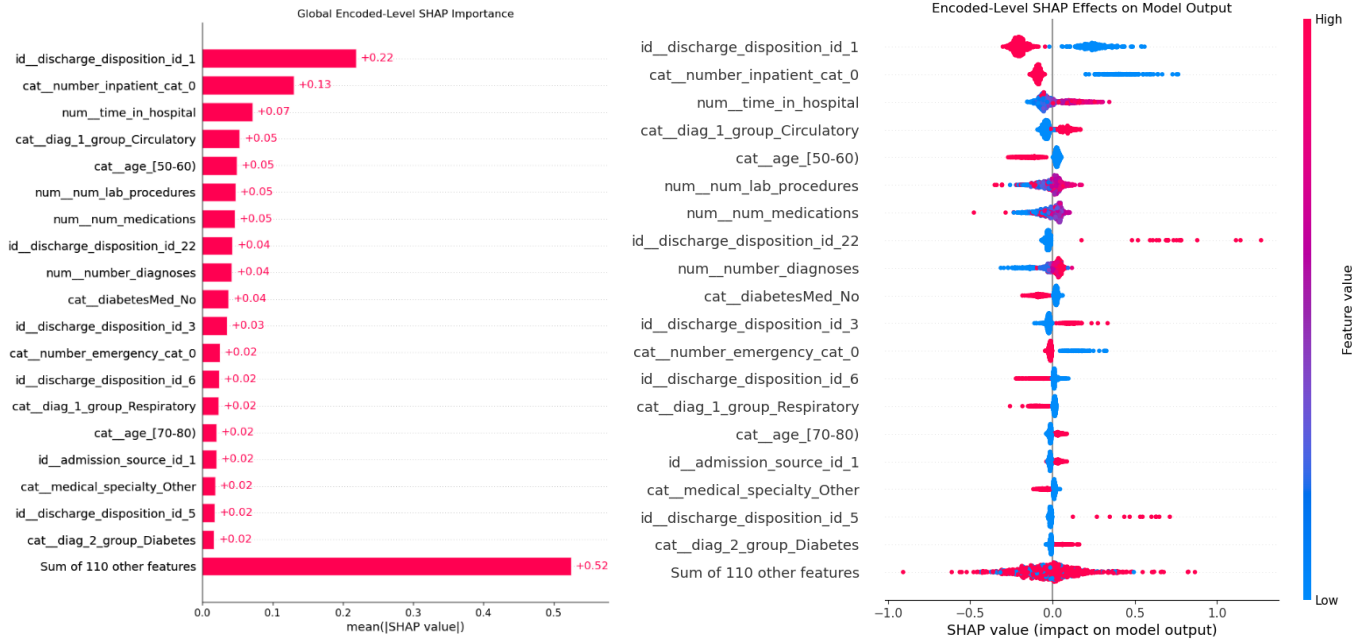


Figure 7. Global encoded-level SHAP importance and SHAP effect distributions for the final XGBoost model. The bar plot reports mean absolute SHAP magnitude (left), while the beeswarm plot shows the direction and distribution of feature contributions on the raw-margin scale (right).

The dominant encoded feature was discharge disposition ID 1 (mean absolute SHAP 0.219, Table 24, Figure 7), which the companion IDS mapping defines as “discharge to home”. The beeswarm plot supplies the direction that the bar chart cannot: when this indicator was present, SHAP values were predominantly negative (Figure 7 right), pushing the model toward lower predicted readmission risk. This was followed by no prior inpatient utilization (0.130), and its presence also generally reduced the model output, whereas having 1 or more prior inpatient encounters increased it. Together, these results show that discharge destination and recent utilization were the model’s most influential risk-stratification signals. Other discharge destinations showed the opposite pattern. Disposition ID 22 means “transfer to a rehabilitation facility”, including a hospital rehabilitation unit. Its presence produced some of the largest positive SHAP values in the sample (Table 24, Figure 7).

Among the remaining leading predictors, longer hospital stays generally increased predicted risk (Table 24, Figure 7). A circulatory primary diagnosis, more recorded diagnoses, and larger numbers of laboratory procedures or medications also tended to increase the model output for many patients, although their effects varied. The age 50-59 indicator generally reduced risk relative to the model’s other encoded age pattern, while age 70-79, which appeared slightly lower in the ranking, generally increased it. Because one-hot features are interpreted relative to all other levels and model interactions, these directions should be read from the complete beeswarm distribution rather than as isolated clinical rules.

The result is clinically plausible as a risk-stratification pattern: the model relies heavily on discharge context, recent utilization, encounter intensity, diagnosis profile, and age. It should not be read as evidence that changing any one of these factors would prevent readmission. Administrative codes may also encode hospital workflow, destination of care, or unmeasured severity.

Rank	Encoded feature level	Mean SHAP
1	Discharged to home (Discharge ID 1)	0.219
2	No prior inpatient utilization	0.130
3	Time in hospital	0.071
4	Primary diagnosis: Circulatory	0.053
5	Age 50-59	0.049
6	Number of laboratory procedures	0.047
7	Number of medications	0.046
8	Transfer to rehab facility (Discharge ID 22)	0.043
9	Number of diagnoses	0.042
10	No diabetes medication	0.037

Table 24. Leading encoded features in the final XGBoost model ranked by mean absolute SHAP value.

Local SHAP explanations were produced for the highest-risk patient in the reproducible 1,000-patient sample (Figure 8) and for a true-positive example (Figure 9). The highest-risk patient had a predicted probability of 0.459, well above the fixed operating threshold of 0.0788. Its SHAP waterfall decomposes the prediction from the model's baseline margin into the encoded feature values that increased or decreased risk. A second waterfall was generated for the first true positive in the sampled order; that patient had a predicted probability of 0.088 and was therefore only slightly above the decision threshold. These examples demonstrate that the same positive classification can arise from very different combinations and strengths of evidence

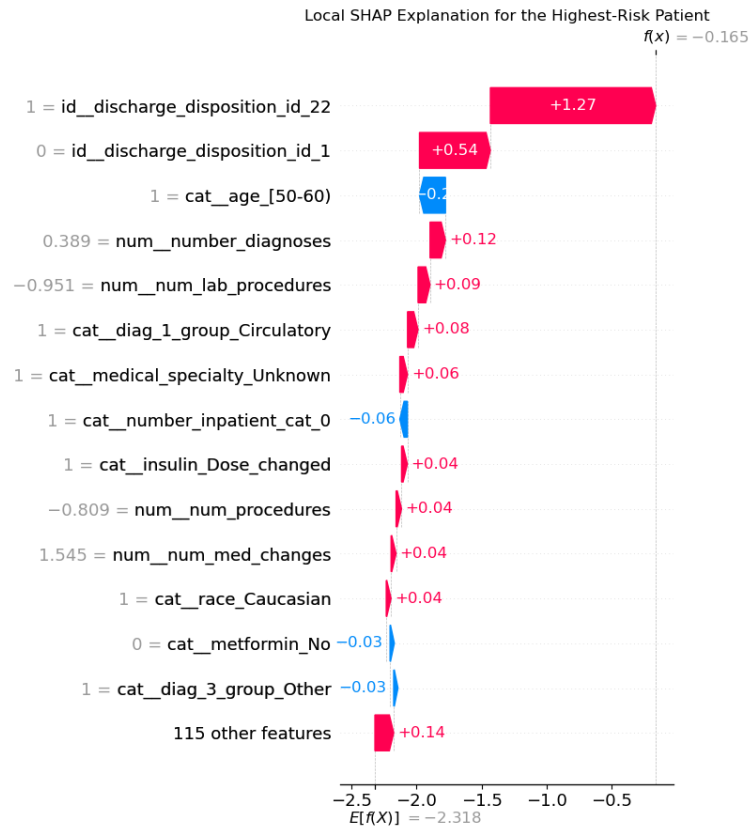


Figure 8. Local SHAP explanation for the highest-risk patient in the 1,000-patient test sample, showing the feature contributions that increased or decreased the model output.

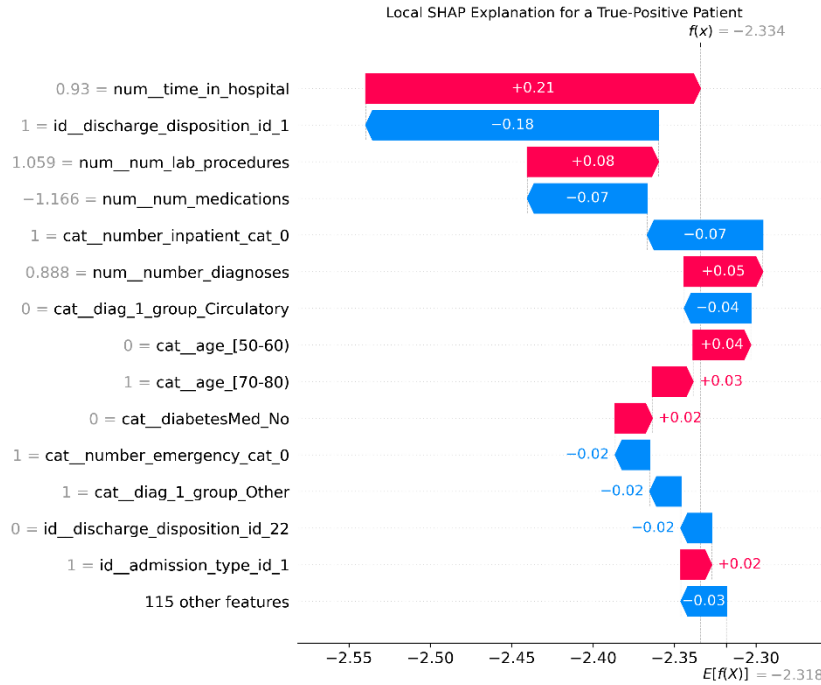


Figure 9. Local SHAP explanation for a true-positive test patient, showing the feature contributions associated with a correctly identified 30-day readmission.

Local explanations are suitable for case review and model governance, but it must be noted that they should not be presented as clinical advice. They are sensitive to the fitted model, background distribution, encoded representation, and correlated features. A reviewer should inspect the original patient values alongside the waterfall, especially when an encoded absence indicator appears important.

8.2 Partial dependence

Time in hospital was the highest-ranked **continuous predictor** (PDP is useful for assessing continuous predictors) and was therefore selected programmatically for partial dependence. The plot was estimated through the complete pipeline on the training set, using predicted probability, 50 grid points, and the 5th-95th percentile range to avoid emphasizing sparse extremes. It shows the model's average learned response over the observed length-of-stay range. It must be noted that the curve is not a causal dose-response (length of stay is related to illness severity, care processes, discharge destination, and other predictors), so varying it while holding all other recorded values fixed may create unrealistic combinations.

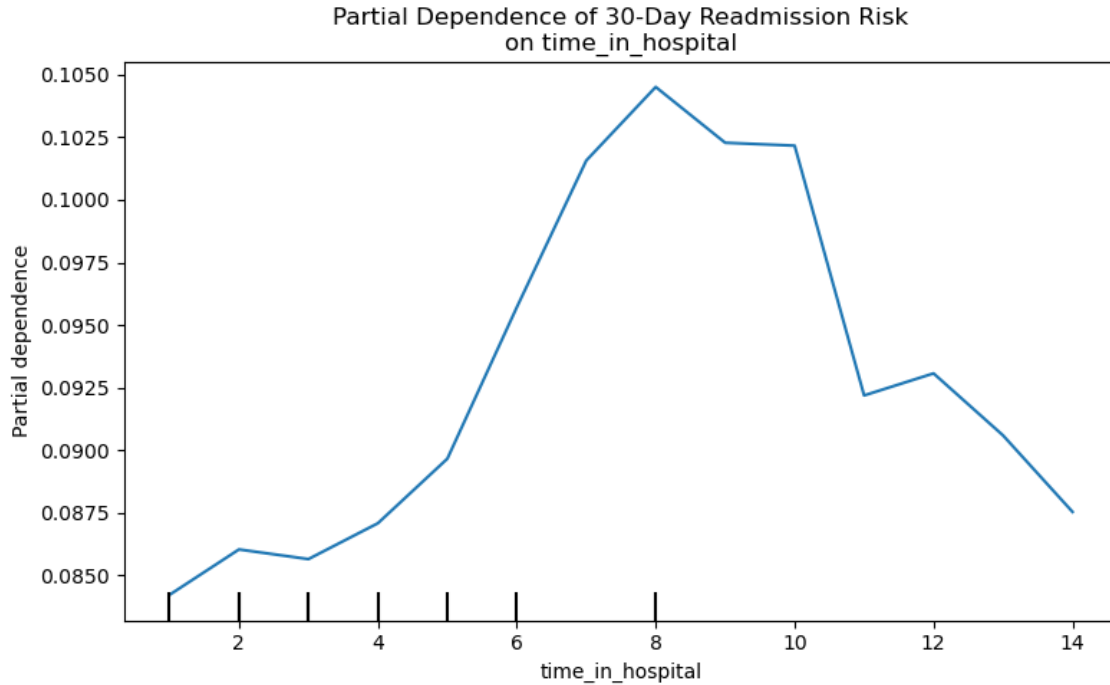


Figure 10. Partial dependence of predicted 30-day readmission risk on time in hospital over the 5th–95th percentile range of the training data.

8.3 Clinical implications and conclusion

The SHAP analysis supports a coherent conclusion: the model is mainly identifying patients whose discharge setting, recent inpatient-use history, and encounter complexity resemble profiles that were historically readmitted within 30 days. The prominence of transfer to rehabilitation, skilled nursing, or another inpatient institution suggests that post-acute care transitions are especially important risk markers. Operationally, these patients may benefit from medication reconciliation, confirmed handover, early follow-up, and coordination across facilities. Patients with prior inpatient use or longer stays may likewise warrant additional transition support. These implications are appropriate for prioritizing supportive outreach, but they do not establish which intervention will prevent readmission.

Discharge disposition is available at the stated prediction time, so its use is not temporal leakage under this discharge-time design. It nevertheless raises important concerns about governance. The variable may encode severity, functional status, social support, insurance constraints, hospital practice, and access to post-acute care. It may also affect where a later admission is recorded. The model should therefore not treat discharge destination as a reason to deny services or as a substitute for clinical assessment. Performance and SHAP direction should be rechecked within each discharge setting and hospital, and a sensitivity model without discharge disposition should be compared to determine how dependent overall and subgroup performance is on this administrative feature.

8.4. Discussions of Limitations/Caveats

The explanations remain post-hoc and inherit the limitations of the historical data. Mean absolute SHAP values measure influence rather than causal effect; one-hot encoding divides importance across category levels; correlated predictors can share or exchange attribution; and the 1,000-patient explanation sample introduces sampling variation. Before deployment, SHAP should be repeated on the complete validation population or several stratified samples, category-level values should be aggregated to their original variables, and explanation stability should be tested across hospitals, periods, discharge destinations, and demographic groups. The practical conclusion is that the model provides a plausible but moderate risk signal for allocating additional support, not evidence that discharge destination or any other feature directly causes readmission.

9. FAIRNESS AND BIAS AUDIT

Because the model is intended to prioritize post-discharge support, unequal selection or error rates could determine which patients receive a beneficial intervention. The audit therefore evaluated race, gender, and age on the untouched test set using the final XGBoost probabilities and the threshold of 0.0788 selected from out-of-fold training predictions. This is a benefit-allocation setting: a false negative may withhold useful support, while a false positive consumes limited capacity but does not imply denial of care.

9.1. Audit design and metrics

For every subgroup, the audit reported sample size, observed prevalence, mean predicted risk, selection rate, recall or true-positive rate (TPR), false-positive rate (FPR), false-negative rate, precision, specificity, Average Precision, ROC-AUC, and Brier loss. Demographic parity difference is the range in selection rates; the disparate-impact ratio is the lowest divided by the highest selection rate; equal-opportunity difference is the TPR range; and equalized-odds difference is the larger of the TPR and FPR ranges. Difference measures are more similar at zero and the ratio is more similar at one. Groups with fewer than 50 observations remained visible in detailed results but were excluded from aggregate gap calculations because their estimates are unstable.

These metrics represent different and sometimes incompatible fairness goals. Demographic parity can be informative for access to a beneficial programme, but unequal underlying readmission prevalence means equal selection rates does not imply equal clinical performance. Equal opportunity focuses on whether actual readmissions are detected similarly, while equalized odds also considers unnecessary flags among non-readmitted patients. The audit therefore uses several measures rather than declaring fairness from a single cutoff such as the commonly cited 0.80 disparate-impact rule.

9.2. Subgroup results

Race: Among groups with at least 50 observations, the demographic-parity difference was 0.108, the disparate-impact ratio was 0.777, the equal-opportunity difference was 0.079, and the equalized-odds difference was 0.140. Selection ranged from 0.375 for Asian patients to 0.483 for Caucasian patients. Recall was comparatively similar, ranging from 0.658 to 0.737, while FPR varied more, from 0.325 to 0.464. Some groups were small (Asian $n=88$, Other $n=231$, Hispanic $n=299$), so their estimates would have been better accompanied with confidence intervals and external validation. Unknown race also represents missing-data processes rather than a coherent demographic group.

Gender: Female and male patients had similar prevalence (0.089 versus 0.091), selection (0.479 versus 0.450), recall (0.684 versus 0.646), and FPR (0.459 versus 0.430). The resulting demographic-parity difference was 0.029, disparate-impact ratio 0.939, and equalized-odds difference 0.037. These relatively small gaps are reassuring within this test set, but the source provides a binary and occasionally invalid gender field and does not support conclusions about gender identity. The single Unknown observation was excluded from gap statistics.

Age: Age showed the largest disparity. For broad age groups, selection rose from 0.236 at ages 0-39 to 0.696 at ages 80+, while recall rose from 0.542 to 0.827 and FPR from 0.218 to 0.682. The demographic-parity difference was 0.460, the disparate-impact ratio 0.339, the equal-opportunity difference 0.329, and the equalized-odds difference 0.464. Some difference reflects higher observed prevalence among older groups (0.100 at 80+ versus 0.055 at 0-39), but prevalence alone does not explain the very large selection and FPR gaps. At the selected low threshold, older non-readmitted patients are much more likely to be flagged, whereas younger true readmissions are more likely to be missed.

Attribute	Group	N	Prev.	Select.	TPR	FPR	Precision
Race	Caucasian	10,509	0.091	0.483	0.665	0.464	0.126
Race	African American	2,504	0.090	0.424	0.658	0.401	0.139
Race	Unknown	364	0.052	0.382	0.737	0.362	0.101
Race	Hispanic	299	0.097	0.391	0.690	0.359	0.171
Race	Other	231	0.048	0.381	0.727	0.364	0.091
Race	Asian	88	0.125	0.375	0.727	0.325	0.242
Gender	Female	7,436	0.089	0.479	0.684	0.459	0.127
Gender	Male	6,558	0.091	0.450	0.646	0.430	0.130
Age	0-39	874	0.055	0.236	0.542	0.218	0.126
Age	40-59	3,811	0.077	0.285	0.498	0.268	0.134
Age	60-79	6,713	0.097	0.508	0.687	0.489	0.132
Age	80+	2,597	0.100	0.696	0.827	0.682	0.119

Table 25. Test-set performance by race, gender, and broad age group at the fixed threshold of 0.0788.

Attribute	DP difference	DI ratio	TPR difference	FPR difference	EO difference
Race	0.108	0.777	0.079	0.140	0.140
Gender	0.029	0.939	0.037	0.029	0.037
Age group	0.460	0.339	0.329	0.464	0.464

Table 26. Aggregate demographic-parity, disparate-impact, equal-opportunity, false-positive-rate, and equalized-odds gaps for the final XGBoost model.

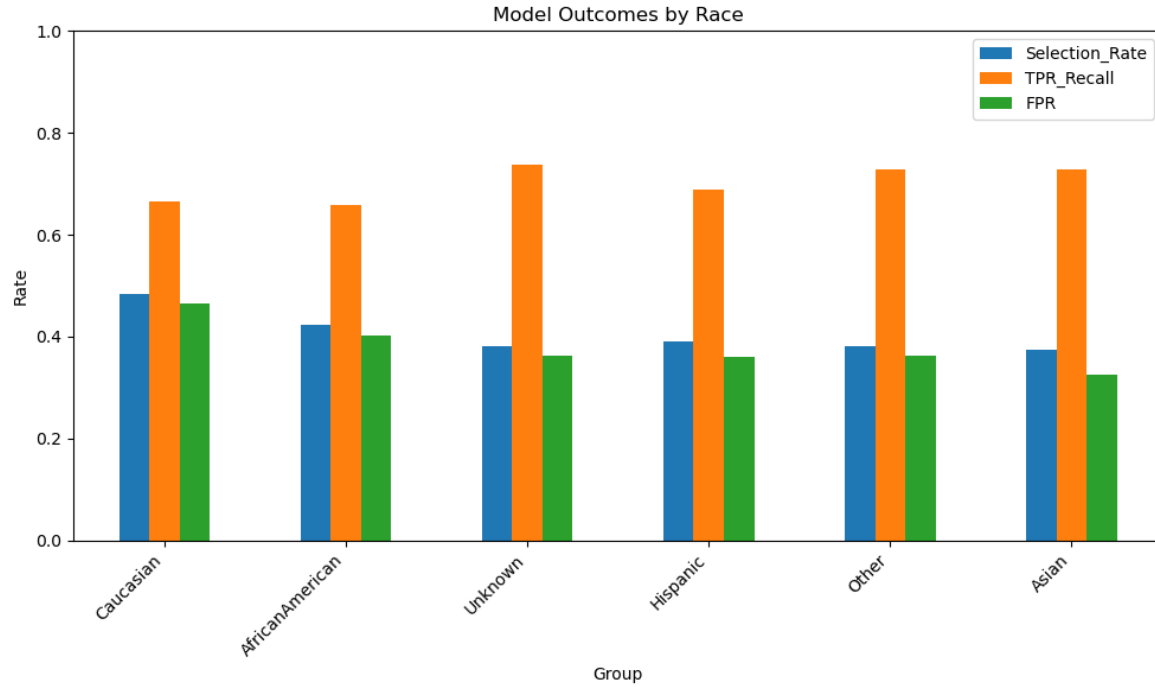


Figure 11. Selection rate, true-positive rate, and false-positive rate by recorded race on the untouched test set.

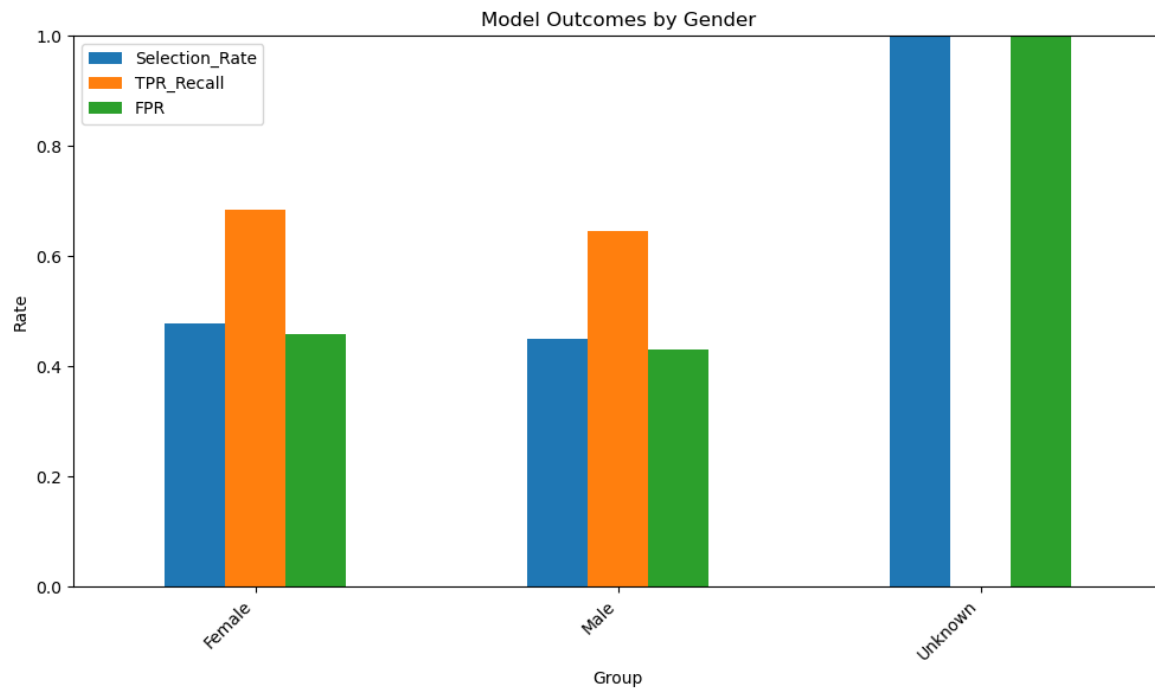


Figure 12. Selection rate, true-positive rate, and false-positive rate by recorded gender on the untouched test set.

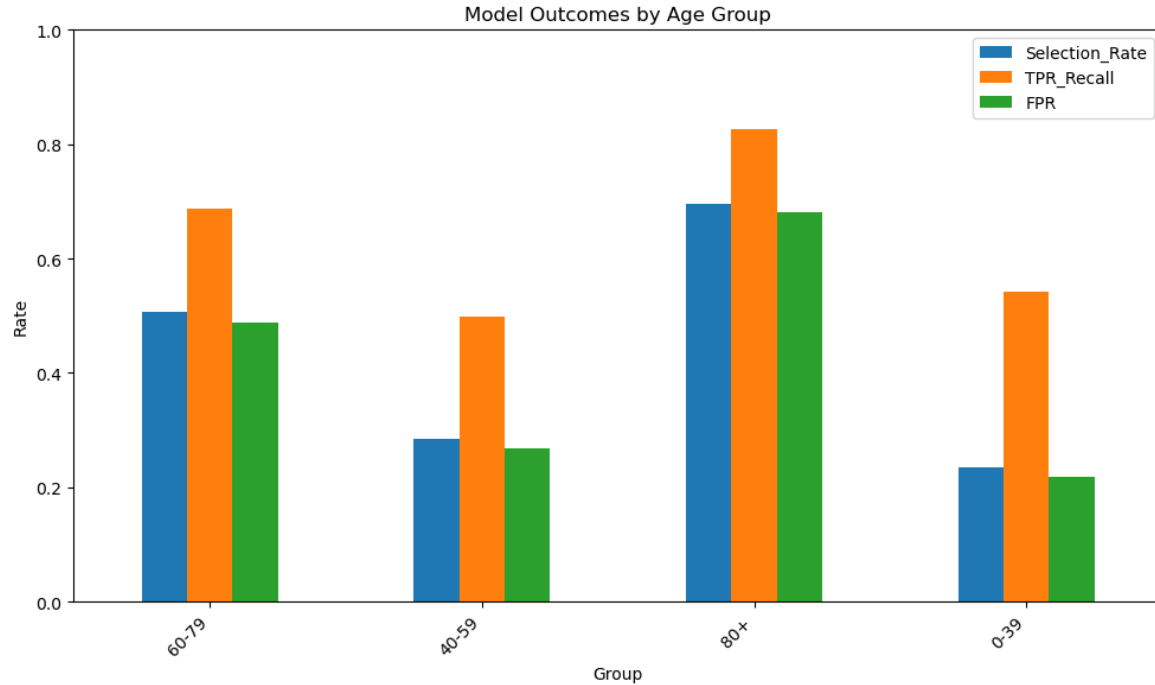


Figure 13. Selection rate, true-positive rate, and false-positive rate by broad age group on the untouched test set.

9.3 Sensitive-attribute removal sensitivity analysis

A secondary XGBoost model used the same selected hyperparameters after race, gender, and age were removed. Its operating threshold was selected independently from out-of-fold training probabilities, so threshold-dependent comparisons reflect both feature removal and a changed threshold. This was a post-hoc sensitivity analysis rather than a test-set-driven replacement-model selection.

Removing the three attributes reduced test Average Precision from 0.183 to 0.177, left ROC-AUC at 0.653 and Brier loss at 0.079, reduced recall from 0.666 to 0.618, and increased specificity from 0.555 to 0.598 (Table 27). It did not consistently reduce disparities. Race equalized-odds difference improved slightly (0.140 to 0.129, Table 27 Blue rows), but race equal-opportunity difference worsened (0.079 to 0.125). Gender gaps increased (Table 28 pink rows). Age equalized-odds difference decreased from 0.464 to 0.426 (Table 28, green rows), yet the age disparate-impact ratio decreased from 0.339 to 0.312. These results demonstrate that fairness through unawareness is insufficient: discharge, utilization, diagnosis, specialty, and other variables can proxy demographic patterns, and subgroup base rates and data-generating processes remain different.

Metric	Original XGBoost	Without race gender age
Average Precision	0.183	0.177
ROC-AUC	0.653	0.653
Recall	0.666	0.618

Metric	Original XGBoost	Without race gender age
Precision	0.128	0.132
Specificity	0.555	0.598
F1	0.215	0.217
Balanced accuracy	0.610	0.608
Brier loss	0.079	0.079

Table 27. Test-set predictive performance of the original XGBoost model and the model trained without race, gender, and age.

Model	Attribute	DP diff.	DI ratio	TPR diff.	FPR diff.	EO diff.
Original	Race	0.108	0.777	0.079	0.140	0.140
Original	Gender	0.029	0.939	0.037	0.029	0.037
Original	Age	0.460	0.339	0.329	0.464	0.464
Attributes removed	Race	0.103	0.762	0.125	0.129	0.129
Attributes removed	Gender	0.041	0.907	0.051	0.041	0.051
Attributes removed	Age	0.421	0.312	0.241	0.426	0.426

Table 28. Fairness gaps before and after removing race, gender, and age from the XGBoost predictors.

9.4 Sources of bias and limitations

The audit does not prove that the model or health system is fair. First, the data cover 1999-2008 care at participating US hospitals and may reflect historical differences in access, treatment, documentation, and follow-up. Second, the outcome is readmission captured by the source system; events at other hospitals may be missed unequally. Third, race and gender are coarse recorded labels, race has missing values, and no socioeconomic variable is available despite its relevance to follow-up care. Fourth, first-encounter selection removes longitudinal information and the cohort excludes death and hospice dispositions. Fifth, subgroup metrics for small racial groups are imprecise, and no intersectional audit was conducted. Finally, equal outcomes cannot be inferred from similar error rates because the downstream intervention's uptake and effectiveness may also vary by group.

9.5 Mitigation and governance recommendations

The model should be used only to offer additional support and never to restrict treatment. Before deployment, it is recommended that the hospital should (1) validate discrimination and calibration prospectively at each site; (2) report bootstrap confidence intervals for subgroup and intersectional metrics; (3) improve collection of race, ethnicity, gender and socioeconomic context with appropriate governance; (4) examine groupwise calibration and intervention uptake; and (5) document the threshold, capacity, intended use, exclusions and escalation route in a model card.

Mitigation should be selected according to the operational harm. If missed readmissions are the priority, threshold optimisation may impose a minimum recall by group while monitoring the resulting FPR and capacity burden. Reweighting or resampling can be evaluated during training, and calibrated post-processing can be tested without changing the original risk ranking. Removing protected attributes alone should not be adopted as the fairness intervention because the sensitivity analysis did not show consistent improvement. Any group-specific threshold policy would require legal, ethical, and clinical review, transparent communication, and prospective evaluation.

Ongoing monitoring should stratify selection, TPR, FPR, precision, calibration and intervention outcomes by race, gender, age, site and time. Triggered review should occur when gaps exceed pre-agreed tolerances, sample sizes become inadequate, coding practices change, or performance drifts. The appropriate final decision is therefore conditional deployment for supportive outreach only, with human oversight and prospective fairness monitoring; the present retrospective audit is evidence for further evaluation, not a declaration that the model is fair.

10. Conclusion

Among the evaluated models, XGBoost provided the strongest discrimination and Average Precision, although the improvement over random forest was modest. On the untouched test set, it identified approximately two-thirds of 30-day readmissions but generated many false-positive alerts, with only 12.8% of flagged patients experiencing the outcome. The model therefore provides moderate ranking value rather than definitive clinical prediction.

SHAP analysis showed that the model relied particularly on discharge destination, previous inpatient utilisation, length of stay, diagnosis profile, and encounter complexity. These variables should be interpreted as risk markers rather than causes of readmission. Fairness analysis identified substantial age-related differences and smaller differences across race and gender groups. Removing the sensitive attributes did not consistently improve fairness, demonstrating that other variables may retain demographic information or reflect underlying differences in care and health status.

Finally, the model should not be deployed as an autonomous decision system. Its most appropriate potential use is to support the allocation of low-risk, beneficial post-discharge services, subject to prospective local validation, capacity-based threshold selection, clinical oversight, subgroup monitoring, and evaluation of whether the intervention itself improves patient outcomes. It should not be used for denying care, reducing services, determining insurance eligibility, replacing clinical judgement or making autonomous clinical decisions.

11. Reproducibility and GitHub Repository

All data-processing, exploratory analysis, feature-engineering, model-development, explainability, and fairness-audit code is available in the project's public GitHub repository:

Repository: <https://github.com/ran-kalamba/capstone-project>

The repository contains the final notebook, documentation, model artefacts, figures, and environment information required to reproduce the analysis. The original public dataset is referenced in the README. The raw data and ID mapping are stored in the data/ directory.

Random seeds were fixed where applicable, including `random_state=42` for the train-test split and reproducible sampling. Preprocessing was fitted using the training data only, and the untouched test set was reserved for final evaluation. The selected operating threshold was derived from out-of-fold training predictions before test-set evaluation.

Item	Location
Main README	README.md
Final report	reports/
Reproducible notebook	notebooks/
Dataset	data/
Saved model and preprocessing pipeline	models/
Generated figures	figures/
Package requirements	requirements.txt
Technical and executive slide decks	slidedecks/

REFERENCES

- Clore, J., Cios, K., DeShazo, J., and Strack, B. (2014). Diabetes 130-US Hospitals for Years 1999-2008. UCI Machine Learning Repository. <https://doi.org/10.24432/C5230J>
- Strack, B., DeShazo, J. P., Gennings, C., Olmo, J. L., Ventura, S., Cios, K. J., and Clore, J. N. (2014). Impact of HbA1c Measurement on Hospital Readmission Rates: Analysis of 70,000 Clinical Database Patient Records. BioMed Research International, 2014, 781670. <https://doi.org/10.1155/2014/781670>

APPENDIX A : SUPPLEMENTARY FIGURES

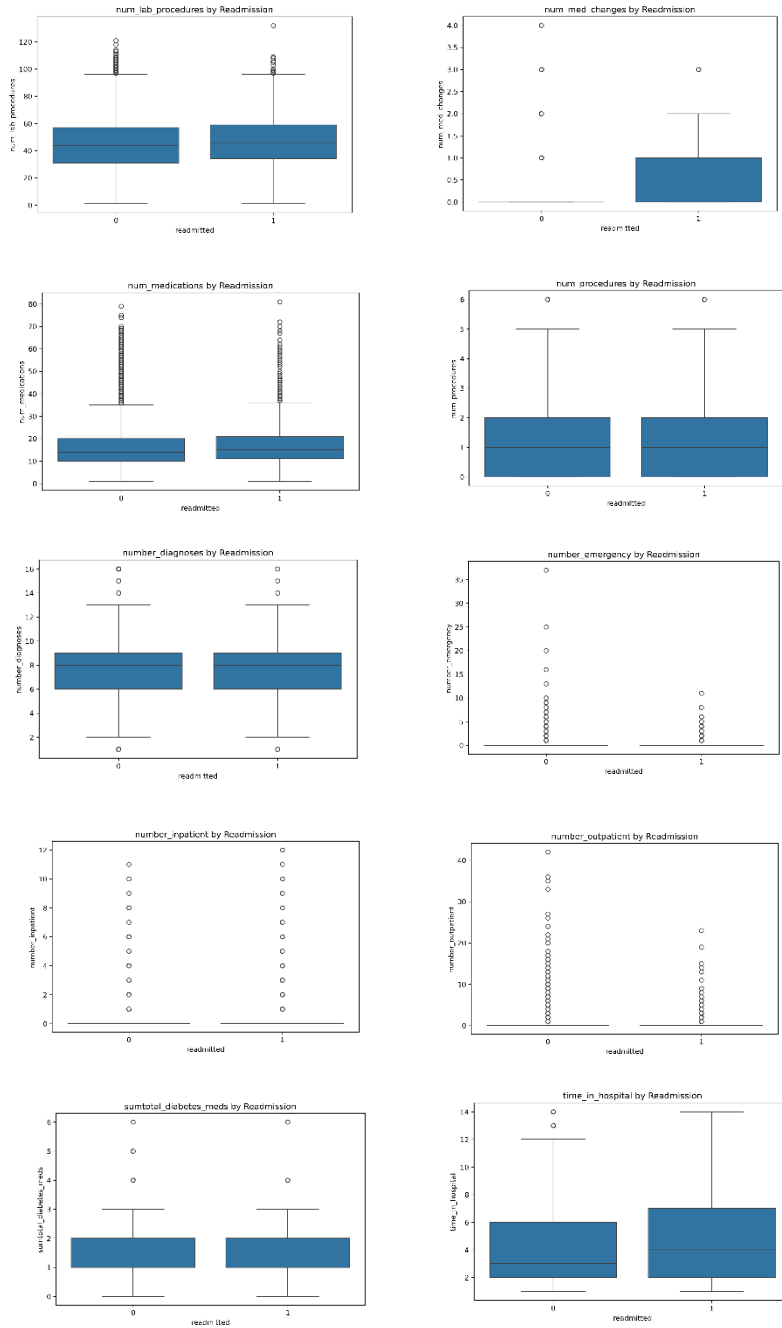


Figure 1A : Boxplots of numeric candidate predictors by 30-day readmission status in the training set.

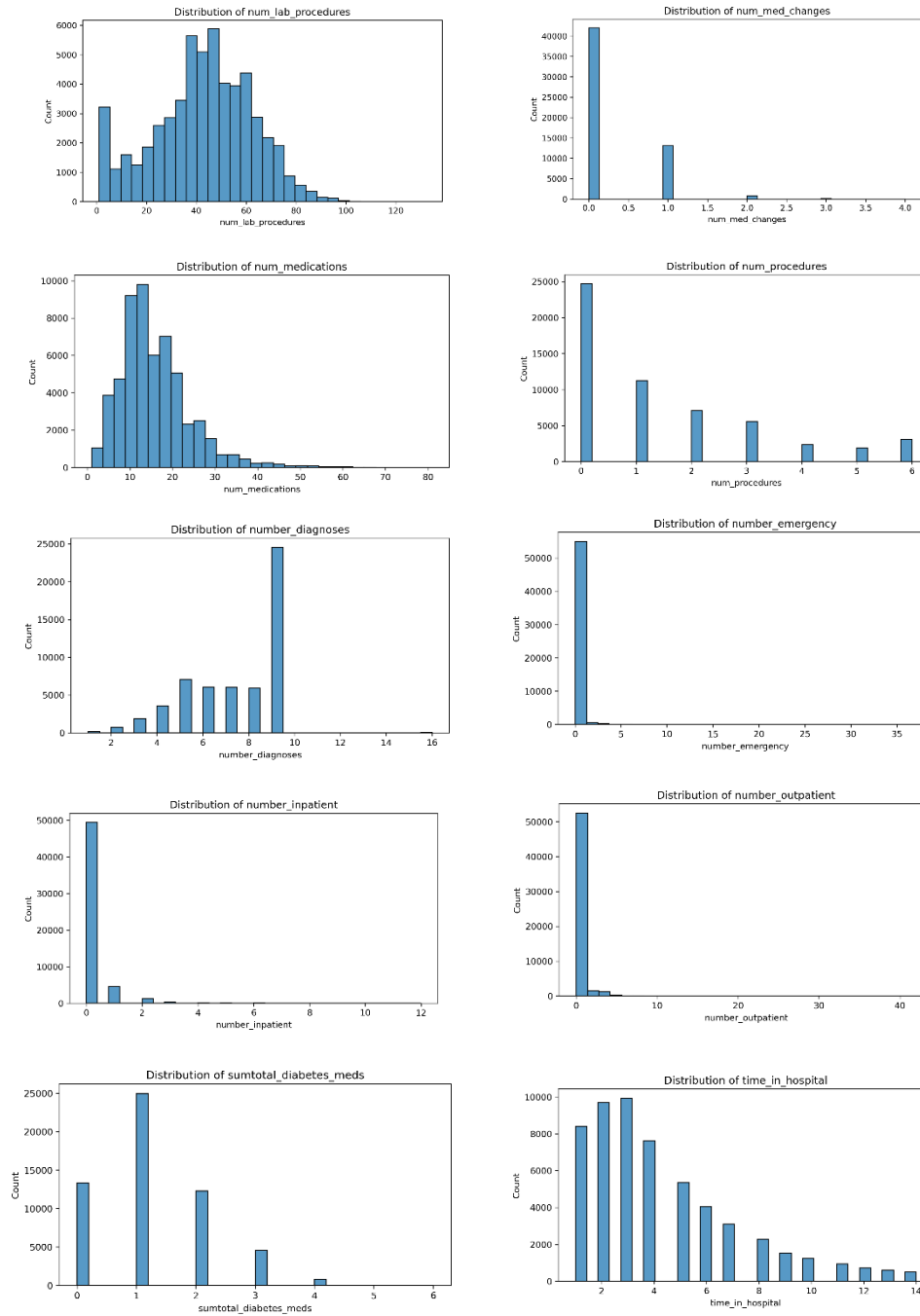


Figure 2A : Training-set distributions of the numeric candidate predictors before final feature engineering.

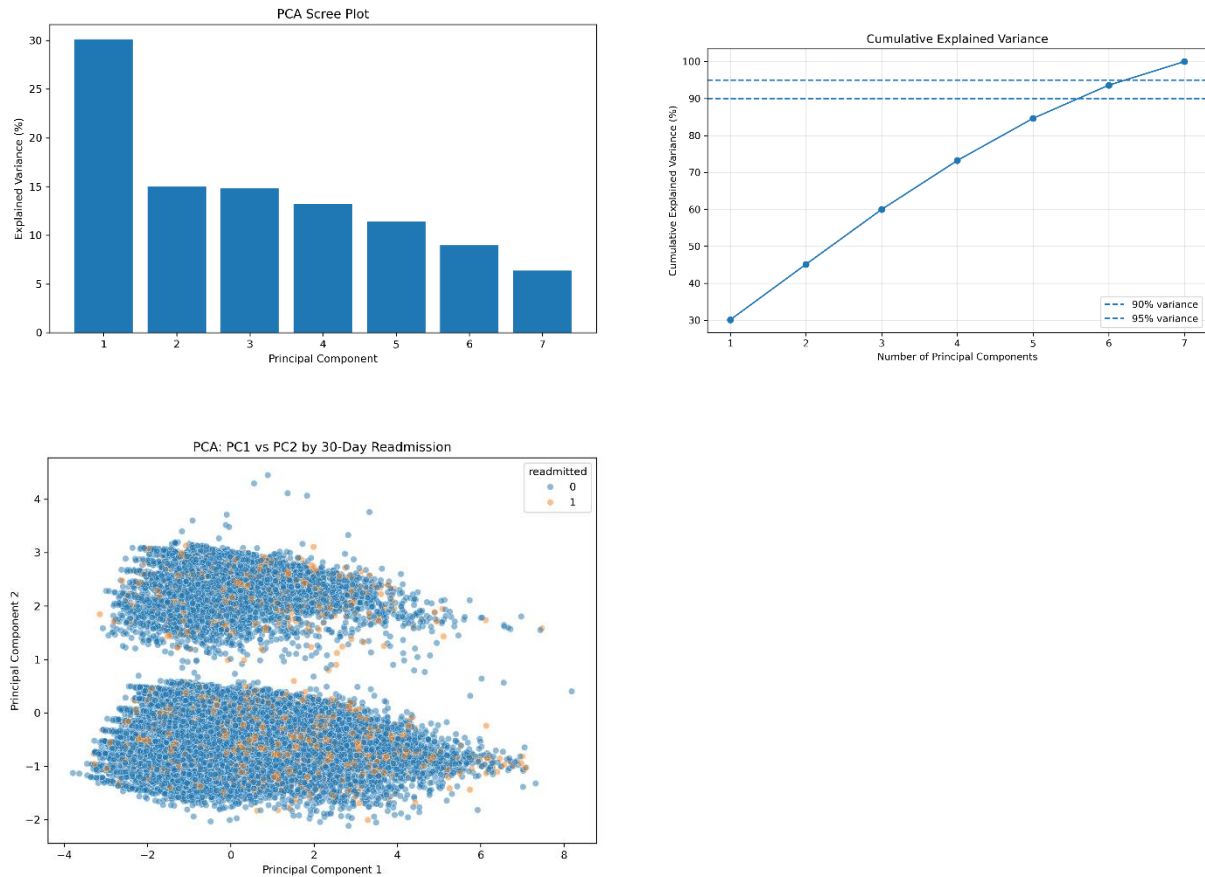


Figure 3A. Principal-component analysis of the seven standardized numeric predictors, including the variance explained and the distribution of observations in principal-component space